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ARTIFICIAL INTELLIGENCE-BASED SYSTEM AND METHOD FOR DETERMINING MEDICAL THERAPIES FOR PATIENTS

Abstract

An AI-based system and method for determining medical therapies is disclosed. The AI-based method includes receiving patient multiomics raw data, extracting one or more reports from the patient multiomics raw data, and detecting one or more genome variants in the patient. Further, the AI-based method includes determining if the one or more genome variants are one or more known variants or one or more unknown variants, determining if the one or more genome variants are one or more coding variants or one or more non-coding variants, and determining if the patient is suffering from a functional loss or a functional excess. The AI-based method includes generating one or more rescue recommendations, determining one or more medical therapies for the patient, and outputting the one or more rescue recommendations and the one or more medical therapies to one or more electronic devices associated with the user.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the priority to incorporates by reference the entire disclosure of U.S. patent application Ser. No. 17/818,034, filed on Aug. 8, 2022, titled “ARTIFICIAL INTELLIGENCE-BASED SYSTEM AND PROCESS FOR PRECISION MEDICINE PRACTICE”.

FIELD OF INVENTION

[0002] Embodiments of the present disclosure relate to Artificial Intelligence (AI)-based systems and more particularly relates to an AI-based system and method for determining one or more medical therapies for a patient.

BACKGROUND

[0003] Treating rare disease is a global challenge. It is often necessary for countries to work together to solve these issues. Rare diseases, which number over 8000 distinct illnesses, collectively affects over 300 million people in the whole world of which 200 million are children. These diseases contribute to the continued mortality of children. For example, rare diseases cause 35% of deaths in the first year of life and 30% of deaths of children before their 5th birthday of those that survive. Currently, the medical field can only diagnose 30% of rare diseases in patients. This conclusion is at odds with the current drug development process as it takes an average of 14 years and 2 billion dollars to develop a new therapeutic for a particular disease. Continued disinterest of pharma and biotech industries to develop therapies result in only 5% of rare disease groups having an FDA-approved treatment. Another related issue is “continued misdiagnosis”. On average, it takes 7.6 years in the U.S. and 5.6 years in the U.K. for correct diagnosis of a rare disease. During this time on average, a patient will often be misdiagnosed 2-3 times and sees four different physicians and four different specialists.

[0004] Recent advances in biomedical sciences have resulted in innovative drug treatments. Yet making these treatments available to consumers remains a challenge. On an average, it takes ten to fifteen years and 1.3 billion dollars to develop one new drug. The process of developing a new drug involves a complex system of testing in which even qualified candidates have a failure rate of 80%. The main reason for such complicated process is that biological systems involve a lot of complicated procedures. Conventionally, first principle-based models, such as molecular dynamics, quantum mechanical, molecular mechanical and the like are used to facilitate a drug discovery process. However, the conventional first principle-based models have multiple disadvantages, such as are treatments problem specific, scale varies, accuracy varies, expensive to compute, and the like. Further, deep learning artificial intelligence models are also used to facilitate the drug discovery process. However, the deep learning artificial intelligence models includes various disadvantages, such as uniformly applicable, over fitting due to data bias, expensive to train, low repeatability, and the like.

[0005] Hence, there is a need for an improved Artificial Intelligence (AI)-based computing system for determining one or more medical therapies corresponding to drug discovery, in order to address the aforementioned issues.

SUMMARY

[0006] This summary is provided to introduce a selection of concepts, in a simple manner, which is further described in the detailed description of the disclosure. This summary is neither intended to

identify key or essential inventive concepts of the subject matter nor to determine the scope of the disclosure.

[0007] In accordance with an embodiment of the present disclosure, an Artificial Intelligence (AI)-based computing system for determining medical therapies for a patient is disclosed. The AI-based computing system includes one or more hardware processors and a memory coupled to the one or more hardware processors. The memory includes a plurality of modules in the form of programmable instructions executable by the one or more hardware processors. The plurality of modules include a patient data receiver module configured to receive patient multiomics raw data associated with the patient from one of: one or more electronic devices associated with a user and an external database. The plurality of modules include a data extracting module configured to extract one or more reports from the received patient multiomics raw data using one or more data extraction techniques. The extracted one or more reports include a patient whole genome report, a patient exome report, and a patient Ribonucleic Acid (RNA) report. Further, the plurality of modules include a variant detection module configured to detect one or more genome variants in the patient by analyzing the extracted patient whole genome report. The one or more genome variants are permanent changes in a Deoxyribonucleic Acid (DNA) sequence that makes up a gene. The plurality of modules also include a variant determination module configured to determine if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants based on the extracted patient whole genome report and one or more diseases of the patient using a variant determination-based Artificial Intelligence (AI) model. The plurality of modules includes a genome determination module configured to determine if the detected one or more genome variants are one of: one or more coding variants and one or more non-coding variants based on the extracted patient whole genome report using a coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants. Further, the plurality of modules includes a functional determination module configured to determine if the patient is suffering from one of: a functional loss and a functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants. The plurality of modules also include a data generation module configured to generate one or more rescue recommendations for the user to perform a functional rescue therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss. Furthermore, the plurality of modules include a medical therapy recommendation module configured to determine one or more medical therapies for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations. The one or more medical therapies correspond to a new drug discovery for one or more potential targets corresponding to the one or more coding variants of the patient. For determining the one or more medical therapies for the patient based on the result of the functional rescue therapy, the medical therapy recommendation module with the functional rescue therapy-based AI model, is configured to: (a) obtain at least one of: the result of the functional rescue therapy from the data generation module, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy; (b) train the functional rescue therapy-based AI model on one or more training datasets associated with one or more known results associated with the functional rescue therapy; (c) extract one or more therapy based features comprising at least one of: molecular signatures of the one or more diseases and therapy response, pathway activation scores, drug-target interaction profiles, and one or more factors corresponding to the patient; (d) analyze the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy; (e) rank the one or more medical therapies based on at least one of: predicted efficacy, safety profiles, and the one or more factors corresponding to the patient; and (f) determine the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue

therapy-based AI model.

[0008] The plurality of modules include a data output module configured to output the generated one or more rescue recommendations and the determined one or more medical therapies on user interface screen of the one or more electronic devices associated with the user.

[0009] In an embodiment, for determining if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants, the variant determination module with the variant determination-based AI model is configured to: (a) obtain information associated with at least one of: the patient whole genome report and the one or more diseases of the patient; (b) extract one or more genome based features from the patient whole genome report, wherein the one or more genome based features comprise at least one of: one or more locations of the one or more genome variants, reference and alternate alleles, and types of the one or more genome variants; (c) train the variant determination-based AI model on the one or more training datasets comprising the one or more known variants associated with the one or more diseases; (d) compare the one or more known variants against one or more known variant databases to analyze genomic context; (e) correlate the analyzed genomic context with information associated with the one or more diseases of the one or more patient; and (f) classify the one or more genome variants as at least one of: the one or more known variants and the one or more unknown variants, based on the correlation of the analyzed genomic context with the information associated with the one or more diseases of the one or more patient.

[0010] In another embodiment, the one or more analysis operations comprises expression level check configured to: (i) analyze the patient RNA report to quantify one or more expression levels of genes affected by the one or more coding variants; and (ii) compare the one or more expression levels of genes to reference ranges for normal function, wherein the functional loss is determined when the one or more expression levels of genes are below to a pre-determined threshold value, and wherein the functional excess is determined when the one or more expression levels of genes exceed the pre-determined threshold value. The one or more analysis operations further comprises sequence-based mutant analysis configured to: (i) determine the patient exome report to analyze specific sequence changes in coding regions; (ii) predict an impact of at least one of: amino acid substitutions, insertions, and deletions, on protein function; and (iii) utilize computational tools to assess whether mutations are caused to at least one of: disrupt protein activity and optimize the protein activity. The one or more analysis operations further comprises structural-based mutant analysis configured to: (i) utilize protein structure prediction tools to model one or more effects of the one or more coding variants on 3D protein structure; (ii) analyze an impact of structural changes to at least one of: binding sites, catalytic domains, and overall protein stability; and (iii) determine whether structural alterations are caused to at least one of: increase and decrease protein function.

[0011] The one or more analysis operations further comprises functional impact analysis configured to: (i) integrate data from at least one of: the patient RNA report and the exome report, to predict overall functional consequences; (ii) analyze one or more factors comprising at least one of: protein domain affected, evolutionary conservation of a mutated region, and known functional effects of similar mutations; and (iii) assess whether the variant is predicted to result in at least one of: loss-of-function and gain-of-function effect on the protein. The one or more analysis operations further comprises pathway analysis configured to: (i) determine whether the affected proteins interact with biological pathways; (ii) determine whether the variant disrupt and over-activate potential cellular processes; and (iii) analyze compensatory mechanisms that at least one of: mitigate and exacerbate the functional changes. The one or more analysis operations further comprises machine learning-based prediction configured to: (i) utilize one or more trained machine learning models to predict functional outcomes based on features extracted from at least one of: genomic and transcriptomic data; and (ii) integrate multiple data types to provide a holistic assessment of functional impact. The one or more analysis operations further comprises literature-

based analysis configured to: (i) automatically search and analyze scientific literature for reported functional effects of similar variants; and (ii) integrate expert knowledge into functional assessment. The one or more analysis operations further comprises experimental data integration configured to: (i) integrate data from at least one of: functional assays and animal models studying similar variants; and (ii) refine predictions of at least one of: the functional loss and the functional excess using the data.

[0012] In yet another embodiment, the one or more rescue recommendations are generated for the user to perform the functional rescue therapy on the patient using one or more techniques comprising at least one of: knowledge-based recommendation techniques, machine learning-based recommendation techniques, rule-based expert techniques, network analysis, structural biology integration techniques, gene therapy recommendation techniques, RNA-based therapy generator, drug repurposing module, combination therapy optimizer, personalized dosing calculator, and therapy simulation engine.

[0013] In yet another embodiment, the AI-based system further comprises a drug discovery module configured to: (a) generate one or more inhibitor recommendations for the user to perform an inhibitor therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional excess, (b) determine if one or more drugs are available for one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations using a target-drug association-based AI model upon generating the one or more inhibitor recommendations; (c) generate one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets, wherein the generated one or more drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user; (d) identify one or more new drugs for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy upon determining that the one or more drugs are not available for the one or more drug targets; (e) determine one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy using a structure-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more drug targets; (f) generate one or more existing drug recommendations for an existing drug repositioning corresponding to the determined one or more existing drugs, wherein the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user.

[0014] For generating the one or more inhibitor recommendations, the drug discovery module is configured to: (i) analyze the result of the one or more analysis operations to identify at least one of: specific molecular pathways and proteins exhibiting excess activity; and (ii) utilize a knowledge base of known inhibitors and one or more targets of the known inhibitors to generate the one or more inhibitor recommendations. For determining if the one or more drugs are available for one or more drug targets corresponding to the one or more coding variants, the drug discovery module is further configured to: (i) integrate data from at least one of: drug databases, clinical trials, and scientific literature; (ii) determine drug-target interactions using one or more machine learning models; and (iii) analyze one or more factors comprising at least one of: binding affinity, off-target effects, and drug related properties to determine whether the one or more drugs are available for one or more drug targets. For generating the one or more drug recommendations for the known drug repositioning corresponding to the determined one or more drugs, the drug discovery module is further configured to: (i) rank the one or more drugs based on the predicted efficacy for one or more drug targets; (ii) analyze potential side effects and contraindications; and (iii) generate one or more reports indicating rationale for each drug recommendation for the known drug repositioning

corresponding to the determined one or more drugs.

[0015] For determining the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets, the drug discovery module with the structure-based drug repurposing engine is configured to: (i) analyze structural similarities between one or more known drug targets and one or more potential drug targets; and (ii) determine one or more indications associated with the one or more existing drugs based on the analyzed structural similarities between the one or more known drug targets and the one or more potential drug targets. For generating the one or more existing drug recommendations for the existing drug repositioning, the drug discovery module is further configured to: (i) prioritize the one or more existing drugs based on the predicted efficacy and safety for the one or more indications; (ii) analyze one or more factors comprising at least one of: dosing, formulation, and delivery for use; and (iii) generate reports indicating the one or more existing drug recommendations for the existing drug repositioning corresponding to the determined one or more existing drugs based on the analyzed one or more factors.

[0016] In yet another embodiment, the AI-based method further comprises a drug repositioning module configured to: (a) receive a patient phenotype report of the patient from the one or more electronic devices associated with the user, wherein the patient phenotype report comprises height, hair color, and DNA sequence of the patient; (b) detect one or more diseases of the patient based on the patient phenotype report and one of: International Classification of Diseases (ICD)-9 and ICD-10 using a phenotype-disease association-based AI model; (c) determine one or more medicines for the detected one or more diseases based on the received patient phenotype report and one or more clinical guidelines using a phenotype-drug association-based AI model upon determining that the detected one or more genome variants are the one or more known variants; and (d) generate one or more medicine recommendations for a known medicine repositioning corresponding to the determined one or more medicines, wherein the generated one or more medicine recommendations are outputted on user interface screen of the one or more electronic devices associated with the user.

[0017] For detecting one or more diseases of the patient, the drug repositioning module with the phenotype-disease association-based AI model configured to: (i) obtain the patient phenotype report comprising at least one of: physical characteristics, clinical symptoms and severity, laboratory test results, medical imaging data, DNA sequence from the patient, and the ICD-9 and ICD-10 codes; (ii) pre-process the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of numerical data, encoding of categorical variables, tokenization and embedding of text data, image pre-processing and augmentation techniques, and sequence encoding for DNA data; (iii) train the phenotype-disease association-based AI model on the one or more training datasets comprising known phenotype-disease associations; (iv) extract one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; (v) generate one or more probability scores for the one or more diseases; and (vi) detect the one or more diseases for each patient by ranking the one or more diseases based on the generated one or more probability scores.

[0018] For determining the one or more medicines for the detected one or more diseases, the drug repositioning module with the phenotype-drug association-based AI model is configured to: (i) obtain the patient phenotype report comprising at least one of: the clinical symptoms and severity, the laboratory test results, medical history, genetic information, and clinical guidelines relevant to patient condition; (ii) pre-process the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of the numerical data, encoding of the categorical variables, and tokenization and embedding of the text data; (iii) train the phenotype-drug association-based AI model on the one or more training datasets comprising known phenotype-drug associations; (iv) extract the one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; (v) generate the one or

more probability scores for the one or more medicines; and (vi) detect the one or more medicines for each patient by ranking the one or more medicines based on the generated one or more probability scores.

[0019] For generating the one or more medicine recommendations for the known medicine repositioning corresponding to the determined one or more medicines, the drug repositioning module is configured to: (i) obtain the one or more medicines determined for the detected one or more diseases; (ii) analyze the known medicine repositioning by performing one or more repositioning analysis techniques comprising at least one of: target-based screening technique, structure-based analysis, the pathway analysis, side-effect based repositioning technique, and literature mining technique; (iii) train the one or more machine learning models on the one or more training datasets comprising one or more successful repositioning cases; (iv) predict the known medicine repositioning corresponding to the determined one or more medicines, based on the trained one or more machine learning modes; and (v) prioritize the one or more medicine recommendations for the known medicine repositioning based on one or more factors comprising at least one of: predicted efficacy, safety profile and potential risks, strength of supporting evidence, and potential market impact and unmet medical needs.

[0020] In yet another embodiment, the data analysis module is configured to: (a) perform a variant biopathway impact analysis on the extracted patient whole genome report upon determining that the one or more detected more genome variants are the one or more non-coding variants; (b) determine if one or more drugs are available for one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a gene-drug association-based AI model upon performing the functional rescue therapy; (c) generate one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets; (d) identify one or more new drugs for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis upon determining that the one or more drugs are not available for the one or more non-coding variants; (e) determine one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a pathway-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more non-coding variants; and (f) generate one or more existing drug recommendations for a known drug repositioning corresponding to the determined one or more existing drugs, wherein the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user.

[0021] In an embodiment, the variant biopathway impact analysis is performed on the extracted patient whole genome report, to assess an impact of the one or more non-coding variants on biological pathways, by: (i) identifying regulatory regions affected by the one or more non-coding variants; (ii) predicting one or more changes in transcription factor binding sites; (iii) assessing potential impacts on gene expression; (iv) mapping affected genes to known biological pathways; and (v) simulating pathway perturbations caused by the one or more non-coding variants. In an embodiment, the data analysis module is configured to determine if one or more drugs are available for one or more drug targets corresponding to the one or more non-coding variants, by: (i) integrating data from one or more drug databases and scientific literature; (ii) analyzing gene-drug associations using a graph neural network; (iii) analyzing at least one of: direct and indirect relationship between one or more drug targets; and (iv) assessing strength and relevance of each potential drug-target association.

[0022] In an embodiment, the data analysis module is configured to generate the one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs, by: (i) ranking drugs based on predicted efficacy for one or more affected pathways; (ii)

assessing potential off-target effects and safety profiles; and (iii) generating explanations for each recommendation for the known drug repositioning corresponding to the determined one or more drugs, wherein the generated one or more drug recommendations are outputted in one or more formats on user interface screen of the one or more electronic devices associated with the user. In an embodiment, the data analysis module is configured to identify the one or more new drugs for the one or more drug targets corresponding to the one or more non-coding variants, by: (i) performing virtual screening of compound libraries against predicted drug targets; and (ii) predicting one or more properties and potential efficacy of the one or more new drugs.

[0023] In an embodiment, the data analysis module is configured to determine the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants, by: (i) mapping drug effects onto the biological pathways; (ii) identifying the one or more existing drugs that compensate for the pathway perturbations; (iii) predicting one or more indications based on pathway-level similarities; and (iv) analyzing drug combinations that synergistically modulate affected pathways. In an embodiment, the data analysis module is configured to generate the one or more existing drug recommendations for a known drug repositioning corresponding to the determined one or more existing drugs, by: (i) prioritizing the one or more existing drugs based on predicted efficacy and safety; and (ii) providing one or more rationales for each recommendation for the known drug repositioning corresponding to the determined one or more existing drugs

[0024] In yet another embodiment, the AI-based computing system further comprises a property determination module configured to: (a) obtain a molecular structure of one or more biomolecules from the user, wherein the molecular structure comprises at least one of: 1-Dimensional (D), 2D, and 3D, and wherein the one or more biomolecules comprise proteins, and nucleic acids; and (b) determine one or more properties of each of the one or more biomolecules based on the obtained molecular structure of the one or more biomolecules using a property determination-based AI model, wherein the one or more properties comprise at least one of: physical properties, chemical properties and biological properties.

[0025] In an embodiment, the property determination module is configured to determine the one or more properties of each of the one or more biomolecules, by: (i) obtaining the molecular structure of the one or more biomolecules from the user; (ii) pre-processing the molecular structure of the one or more biomolecules to generate pre-processed molecular structure, by: converting the molecular structure into one or more graph representations; generating molecular fingerprints; sequence encoding for proteins and nucleic acids; and normalizing numerical property data; (iii) training the property determination-based AI model on the one or more training datasets comprising one or more known properties; (iv) extracting one or more physics-based features comprising at least one of electronic properties and conformational energies, to generate one or more learned embeddings comprising one or more molecular characteristics; (v) processing the molecular structure and sequence data through the trained property determination-based AI model to generate unified representation of the one or more biomolecules; and (vi) determining the one or more properties of each of the one or more biomolecules based on the unified representation of the one or more biomolecules.

[0026] In yet another embodiment, the AI-based computing system further comprises a molecule determination module configured to: (a) obtain one or more desired properties of one or more biomolecules from the user, wherein the one or more desired properties comprise at least one of: desired physical properties, desired chemical properties and desired biological properties; and (b) determine one or more novel biomolecules based on the obtained one or more desired properties using a molecule determination-based AI model.

[0027] In an embodiment, the molecule determination module is configured to determine the one or more novel biomolecules based on the obtained one or more desired properties, by: (i) obtaining the one or more desired properties of the one or more biomolecules from the user; (ii) performing a

data representation process on the one or more desired properties of the one or more biomolecules; (iii) training the molecule determination-based AI model on the one or more training datasets comprising the structure-property relationships using one or more known molecules; (iv) generating the one or more biomolecules with the one or more desired properties using the trained molecule determination-based AI model; and (v) processing the one or more biomolecules with the one or more desired properties, by at least one of: encoding the one or more desired properties into a latent space representation; generating candidate biomolecules based on the one or more encoded properties, using a generative model; evaluating the candidate molecules using one or more integrated property prediction models; continuously refining the candidate modules to match the one or more desired properties; and applying one or more chemical validity checks and synthetic accessibility assessments to determine the one or more novel biomolecules.

[0028] In accordance with another embodiment of the present disclosure, an AI-based method for determining medical therapies for a patient is disclosed. The AI-based method includes receiving patient multiomics raw data associated with a patient from one of: one or more electronic devices associated with a user and an external database. The AI-based method includes extracting one or more reports from the received patient multiomics raw data using one or more data extraction techniques. The extracted one or more reports comprise a patient whole genome report, a patient exome report, and a patient RNA report. Further, the AI-based method includes detecting one or more genome variants in the patient by analyzing the extracted patient whole genome report. The one or more genome variants are permanent changes in a DNA sequence that makes up a gene. The AI-based method also includes determining if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants based on the extracted patient whole genome report and one or more diseases of the patient using a variant determination-based AI model. The AI-based method further includes determining if the detected one or more genome variants are one of: one or more coding variants and one or more non-coding variants based on the extracted patient whole genome report using a coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants. Further, the AI-based method includes determining if the patient is suffering from one of: a functional loss and a functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants. Furthermore, the AI-based method includes generating one or more rescue recommendations for the user to perform a functional rescue therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss. The AI-based method includes determining one or more medical therapies for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations. In an embodiment, determining the one or more medical therapies for the patient based on the result of the functional rescue therapy, comprises: (a) obtaining, by the one or more hardware processors, at least one of: the result of the functional rescue therapy from the data generation module, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy; (b) training, by the one or more hardware processors, the functional rescue therapy-based AI model on one or more training datasets associated with one or more known results associated with the functional rescue therapy; (c) extracting, by the one or more hardware processors, one or more therapy based features comprising at least one of: molecular signatures of the one or more diseases and therapy response, pathway activation scores, drug-target interaction profiles, and one or more factors corresponding to the patient; (d) analyzing, by the one or more hardware processors, the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy; (e) ranking, by the one or more hardware processors, the one or more medical therapies based on at least one of: predicted efficacy, safety profiles, and the one or more factors corresponding to the patient; and (f) determining, by the one or more hardware processors,

the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue therapy-based AI model.

[0029] The one or more medical therapies correspond to a new drug discovery for one or more potential targets corresponding to the one or more coding variants of the patient. The AI-based method also includes outputting the generated one or more rescue recommendations and the determined one or more medical therapies on user interface screen of the one or more electronic devices associated with the user.

[0030] Embodiment of the present disclosure also provide a non-transitory computer-readable storage medium having instructions stored therein that, when executed by a hardware processor, cause the processor to perform method steps as described above.

[0031] To further clarify the advantages and features of the present disclosure, a more particular description of the disclosure will follow by reference to specific embodiments thereof, which are illustrated in the appended figures. It is to be appreciated that these figures depict only typical embodiments of the disclosure and are therefore not to be considered limiting in scope. The disclosure will be described and explained with additional specificity and detail with the appended figures.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0032] The disclosure will be described and explained with additional specificity and detail with the accompanying figures in which:

[0033] FIG. 1 is a block diagram illustrating an exemplary computing environment for determining one or more medical therapies, in accordance with an embodiment of the present disclosure;

[0034] FIG. 2 is a block diagram illustrating an exemplary Artificial Intelligence (AI)-based computing system for determining the one or more medical therapies, in accordance with an embodiment of the present disclosure;

[0035] FIG. 3 is a block diagram depicting an exemplary operation of the AI-based computing system, in accordance with an embodiment of the present disclosure;

[0036] FIG. 4 is a block diagram illustrating the AI-based computing system for determining the one or more medical therapies, in accordance with another embodiment of the present disclosure; and

[0037] FIG. 5 is a process flow diagram illustrating an exemplary AI-based method for determining the one or more medical therapies, in accordance with an embodiment of the present disclosure.

[0038] Further, those skilled in the art will appreciate that elements in the figures are illustrated for simplicity and may not have necessarily been drawn to scale. Furthermore, in terms of the construction of the device, one or more components of the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the figures with details that will be readily apparent to those skilled in the art having the benefit of the description herein.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0039] For the purpose of promoting an understanding of the principles of the disclosure, reference will now be made to the embodiment illustrated in the figures and specific language will be used to describe them. It will nevertheless be understood that no limitation of the scope of the disclosure is thereby intended. Such alterations and further modifications in the illustrated system, and such further applications of the principles of the disclosure as would normally occur to those skilled in the art are to be construed as being within the scope of the present disclosure. It will be understood by those skilled in the art that the foregoing general description and the following detailed

description are exemplary and explanatory of the disclosure and are not intended to be restrictive thereof.

[0040] In the present document, the word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or implementation of the present subject matter described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

[0041] The terms “comprise”, “comprising”, or any other variations thereof, are intended to cover a non-exclusive inclusion, such that one or more devices or sub-systems or elements or structures or components preceded by “comprises . . . a” does not, without more constraints, preclude the existence of other devices, sub-systems, additional sub-modules. Appearances of the phrase “in an embodiment”, “in another embodiment” and similar language throughout this specification may, but not necessarily do, all refer to the same embodiment.

[0042] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art to which this disclosure belongs. The system, methods, and examples provided herein are only illustrative and not intended to be limiting.

[0043] A computer system (standalone, client or server computer system) configured by an application may constitute a “module” (or “subsystem”) that is configured and operated to perform certain operations. In one embodiment, the “module” or “subsystem” may be implemented mechanically or electronically, so a module include dedicated circuitry or logic that is permanently configured (within a special-purpose processor) to perform certain operations. In another embodiment, a “module” or “subsystem” may also comprise programmable logic or circuitry (as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations.

[0044] Accordingly, the term “module” or “subsystem” should be understood to encompass a tangible entity, be that an entity that is physically constructed permanently configured (hardwired) or temporarily configured (programmed) to operate in a certain manner and/or to perform certain operations described herein.

[0045] Referring now to the drawings, and more particularly to FIGS. **1** through FIG. **5**, where similar reference characters denote corresponding features consistently throughout the figures, there are shown preferred embodiments and these embodiments are described in the context of the following exemplary system and/or method.

[0046] FIG. **1** is a block diagram illustrating an exemplary computing environment **100** for determining one or more medical therapies, in accordance with an embodiment of the present disclosure. According to FIG. **1**, the computing environment **100** includes one or more electronic devices **102** associated with a user communicatively coupled to an Artificial intelligence (AI)-based computing system **104** via a network **106**. In an exemplary embodiment of the present disclosure, the user may include a medical professional performing a process of determining the one or more medical therapies correspond to a drug discovery. In an embodiment of the present disclosure, drug discovery is a process which aims at identifying a compound therapeutically useful in curing and treating disease. Further, the one or more electronic devices **102** are used by the user for providing patient multiomics raw data associated with a patient to the AI-based computing system **104**. Furthermore, the one or more electronic devices **102** may also be used by the user to receive generated one or more rescue recommendations and determined one or more medical therapies from the AI-based computing system **104**. The AI-based computing system **104** may be hosted on a central server, such as cloud server or a remote server. In an embodiment of the present disclosure, the AI-based computing system **104** may be a cloud-based virtual screening platform for drug candidate screening. Further, the network **106** may be a Wireless-Fidelity (Wi-Fi) connection, a hotspot connection, a Bluetooth connection, a local area network, a wide area network or any other wireless network. In an exemplary embodiment of the present disclosure, the one or more

electronic devices **102** may include a laptop computer, desktop computer, tablet computer, smartphone, wearable device, smart watch, and the like.

[0047] Further, the computing environment **100** includes an external database **108** communicatively coupled to the AI-based computing system **104** via the network **106**. The external database **108** stores the patient multiomics raw data. In an embodiment of the present disclosure, the external database **108** provides the patient multiomics raw data to the AI-based computing system **104**.

[0048] Furthermore, the one or more electronic devices **102** include a local browser, a mobile application or a combination thereof. Furthermore, the user may use a web application via the local browser, the mobile application, or a combination thereof to communicate with the AI-based computing system **104**. In an embodiment of the present disclosure, the AI-based computing system **104** includes a plurality of modules **110**. Details on the plurality of modules **110** have been elaborated in subsequent paragraphs of the present description with reference to FIG. 2.

[0049] In an embodiment of the present disclosure, the AI-based computing system **104** is configured to receive the patient multiomics raw data associated with the patient from the one or more electronic devices **102** associated with the user or the external database **108**. The AI-based computing system **104** extracts one or more reports from the received patient multiomics raw data using one or more data extraction techniques. In an embodiment of the present disclosure, the extracted one or more reports include a patient whole genome report, a patient exome report, and a patient Ribonucleic Acid (RNA) report. The AI-based computing system **104** detects one or more genome variants in the patient by analyzing the extracted patient whole genome report. Further, the AI-based computing system **104** determines if the detected one or more genome variants are one or more known variants, or one or more unknown variants based on the extracted patient whole genome report and one or more diseases of the patient using a variant determination-based Artificial Intelligence (AI) model. The AI-based computing system **104** determines if the detected one or more genome variants are one or more coding variants, or one or more non-coding variants based on the extracted patient whole genome report using a coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants. The AI-based computing system **104** determines if the patient is suffering from a functional loss or a functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants. The AI-based computing system **104** generates one or more rescue recommendations for the user to perform a functional rescue therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss. Furthermore, the AI-based computing system **104** determines one or more medical therapies for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations. In an embodiment of the present disclosure, the one or more medical therapies correspond to a new drug discovery for one or more potential targets corresponding to the one or more coding variants of the patient.

[0050] In an embodiment, for determining the one or more medical therapies for the patient based on the result of the functional rescue therapy, the AI-based computing system **104** is configured to obtain at least one of: the result of the functional rescue therapy, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy. The AI-based computing system **104** is further configured to train the functional rescue therapy-based AI model on one or more training datasets associated with one or more known results associated with the functional rescue therapy. The AI-based computing system **104** is further configured to extract one or more therapy based features comprising at least one of: molecular signatures of the one or more diseases and therapy response, pathway activation scores, drug-target interaction profiles, and one or more factors corresponding to the patient. The AI-based computing system **104** is further configured to

analyze the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy. The AI-based computing system **104** is further configured to rank the one or more medical therapies based on at least one of: predicted efficacy, safety profiles, and the one or more factors corresponding to the patient. The AI-based computing system **104** is further configured to determine the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue therapy-based AI model.

[0051] The AI-based computing system **104** outputs the generated one or more rescue recommendations and the determined one or more medical therapies on user interface screen of the one or more electronic devices **102** associated with the user.

[0052] FIG. **2** is a block diagram illustrating an exemplary AI-based computing system **104** for drug discovery, in accordance with an embodiment of the present disclosure. Further, the AI-based computing system **104** includes one or more hardware processors **202**, a memory **204** and a storage unit **206**. The one or more hardware processors **202**, the memory **204** and the storage unit **206** are communicatively coupled through a system bus **208** or any similar mechanism. The memory **204** comprises the plurality of modules **110** in the form of programmable instructions executable by the one or more hardware processors **202**. Further, the plurality of modules **110** includes a patient data receiver module **210**, a data extracting module **212**, a variant detection module **214**, a variant determination module **216**, a genome determination module **218**, a functional determination module **220**, a data generation module **222**, a medical therapy recommendation module **224**, a data output module **226**, a drug discovery module **228**, a drug repositioning module **230**, a data analysis module **232**, a property determination module **234**, and a molecule determination module **236**.

[0053] The one or more hardware processors **202**, as used herein, means any type of computational circuit, such as, but not limited to, a microprocessor unit, microcontroller, complex instruction set computing microprocessor unit, reduced instruction set computing microprocessor unit, very long instruction word microprocessor unit, explicitly parallel instruction computing microprocessor unit, graphics processing unit, digital signal processing unit, or any other type of processing circuit. The one or more hardware processors **202** may also include embedded controllers, such as generic or programmable logic devices or arrays, application specific integrated circuits, single-chip computers, and the like.

[0054] The memory **204** may be non-transitory volatile memory and non-volatile memory. The memory **204** may be coupled for communication with the one or more hardware processors **202**, such as being a computer-readable storage medium. The one or more hardware processors **202** may execute machine-readable instructions and/or source code stored in the memory **204**. A variety of machine-readable instructions may be stored in and accessed from the memory **204**. The memory **204** may include any suitable elements for storing data and machine-readable instructions, such as read only memory, random access memory, erasable programmable read only memory, electrically erasable programmable read only memory, a hard drive, a removable media drive for handling compact disks, digital video disks, diskettes, magnetic tape cartridges, memory cards, and the like. In the present embodiment, the memory **204** includes the plurality of modules **110** stored in the form of machine-readable instructions on any of the above-mentioned storage media and may be in communication with and executed by the one or more hardware processors **202**.

[0055] The storage unit **206** may be a cloud storage, a Structured Query Language (SQL) data store or a location on a file system directly accessible by the plurality of modules **110**. The storage unit **206** may store the patient multiomics raw data, the one or more reports, the one or more genome variants, the one or more rescue recommendations, the one or more medical therapies, one or more inhibitor recommendations, one or more drug recommendations, one or more new drugs, one or more existing drugs, a patient phenotype report, one or more diseases of the patient, one or more medicines, one or more medicine recommendations, a molecular structure of one or more biomolecules, one or more properties of each of the one or more biomolecules, one or more desired

properties, one or more novel biomolecules, and the like.

[0056] The patient data receiver module **210** is configured to receive the patient multiomics raw data associated with the patient from the one or more electronic devices **102** associated with the user or the external database **108**. In an exemplary embodiment of the present disclosure, the user may include the medical professional performing the determination of the one or more medical therapies correspond to the drug discovery process. In an embodiment of the present disclosure, the drug discovery is a process which aims at identifying the compound therapeutically useful in curing and treating disease. In an exemplary embodiment of the present disclosure, the one or more electronic devices **102** may include a laptop computer, desktop computer, tablet computer, smartphone, wearable device, smart watch, and the like.

[0057] The data extracting module **212** is configured to extract the one or more reports from the received patient multiomics raw data using the one or more data extraction techniques. In an exemplary embodiment of the present disclosure, the extracted one or more reports include a patient whole genome report, a patient exome report, and a patient Ribonucleic Acid (RNA) report.

[0058] The variant detection module **214** is configured to detect the one or more genome variants in the patient by analyzing the extracted patient whole genome report. In an embodiment of the present disclosure, the one or more genome variants are permanent changes in a Deoxyribonucleic Acid (DNA) sequence that makes up a gene.

[0059] The variant determination module **216** is configured to determine if the detected one or more genome variants are the one or more known variants, or the one or more unknown variants based on the extracted patient whole genome report and the one or more diseases of the patient using the variant determination-based Artificial Intelligence (AI) model. In an embodiment, for determining if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants, the variant determination module **216** with the variant determination-based AI model is configured to obtain information associated with at least one of: the patient whole genome report and the one or more diseases of the patient. The variant determination module **216** with the variant determination-based AI model is further configured to extract one or more genome based features from the patient whole genome report, wherein the one or more genome based features comprise at least one of: one or more locations of the one or more genome variants, reference and alternate alleles, and types of the one or more genome variants.

[0060] The variant determination module **216** with the variant determination-based AI model is further configured to train the variant determination-based AI model on the one or more training datasets comprising the one or more known variants associated with the one or more diseases. The variant determination module **216** with the variant determination-based AI model is further configured to compare the one or more known variants against one or more known variant databases to analyze genomic context. The variant determination module **216** with the variant determination-based AI model is further configured to correlate the analyzed genomic context with information associated with the one or more diseases of the one or more patient. The variant determination module **216** with the variant determination-based AI model is further configured to classify the one or more genome variants as at least one of: the one or more known variants and the one or more unknown variants, based on the correlation of the analyzed genomic context with the information associated with the one or more diseases of the one or more patient.

[0061] The genome determination module **218** is configured to determine if the detected one or more genome variants are the one or more coding variants, or the one or more non-coding variants based on the extracted patient whole genome report using the coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants. In an embodiment, for determining if the detected one or more genome variants are at least one of: the one or more coding variants, and the one or more non-coding variants, the genome determination module **218** with the coding determination-based AI model is configured to obtain the one or more unknown variants from the patient whole genome report. The genome

determination module **218** with the coding determination-based AI model is further configured to train the coding determination-based AI model on the one or more training datasets comprising the one or more known coding and non-coding variants. The training of the coding determination-based AI model may involve data augmentation techniques to increase diversity and transfer learning from pre-trained genomic models. In an embodiment, the coding determination-based AI model may be a convolutional neural network (CNN) model. The genome determination module **218** with the coding determination-based AI model is further configured to process one or more features through the CNN model and fully connected layers to generate one or more probability scores for the one or more genome variants to classify at least one of: the one or more coding variants and the one or more non-coding variants.

[0062] The functional determination module **220** is configured to determine if the patient is suffering from the functional loss or the functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants. In an embodiment of the present disclosure, the one or more analysis operations include an expression level check, a sequence based mutant analysis, a structural based mutant analysis, a functional impact analysis, or a combination thereof. In an embodiment of the present disclosure, the sequence based mutant analysis, the structure based mutant analysis, and functional impact analysis are performed on the patient exome report, and the expression level check and the functional impact analysis is performed on the patient RNA report to determine if the patient is suffering from the functional loss or the functional excess.

[0063] In an embodiment, the one or more analysis operations comprises the expression level check that is configured to: (a) analyze the patient RNA report to quantify one or more expression levels of genes affected by the one or more coding variants; and (b) compare the one or more expression levels of genes to reference ranges for normal function, wherein the functional loss is determined when the one or more expression levels of genes are below to a pre-determined threshold value, and wherein the functional excess is determined when the one or more expression levels of genes exceed the pre-determined threshold value.

[0064] The one or more analysis operations further comprises the sequence-based mutant analysis configured to: (a) determine the patient exome report to analyze specific sequence changes in coding regions; (b) predict an impact of at least one of: amino acid substitutions, insertions, and deletions, on protein function; and (c) utilize computational tools to assess whether mutations are caused to at least one of: disrupt protein activity and optimize the protein activity. The one or more analysis operations further comprises structural-based mutant analysis configured to: (a) utilize protein structure prediction tools to model one or more effects of the one or more coding variants on 3D protein structure; (b) analyze an impact of structural changes to at least one of: binding sites, catalytic domains, and overall protein stability; and (c) determine whether structural alterations are caused to at least one of: increase and decrease protein function.

[0065] The one or more analysis operations further comprises functional impact analysis configured to: (a) integrate data from at least one of: the patient RNA report and the exome report, to predict overall functional consequences; (b) analyze one or more factors comprising at least one of: protein domain affected, evolutionary conservation of a mutated region, and known functional effects of similar mutations; and (c) assess whether the variant is predicted to result in at least one of: loss-of-function and gain-of-function effect on the protein. The one or more analysis operations further comprises pathway analysis configured to: (a) determine whether the affected proteins interact with biological pathways; (b) determine whether the variant disrupt and over-activate potential cellular processes; and (c) analyze compensatory mechanisms that at least one of: mitigate and exacerbate the functional changes.

[0066] The one or more analysis operations further comprises machine learning-based prediction configured to: (a) utilize one or more trained machine learning models to predict functional outcomes based on features extracted from at least one of: genomic and transcriptomic data; and

(b) integrate multiple data types to provide a holistic assessment of functional impact. The one or more analysis operations further comprises literature-based analysis configured to: (a) automatically search and analyze scientific literature for reported functional effects of similar variants; and (b) integrate expert knowledge into functional assessment. The one or more analysis operations further comprises experimental data integration configured to: (a) integrate data from at least one of: functional assays and animal models studying similar variants; and (b) refine predictions of at least one of: the functional loss and the functional excess using the data.

[0067] The data generation module **222** is configured to generate the one or more rescue recommendations for the user to perform the functional rescue therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss. In an embodiment, the one or more rescue recommendations are generated for the user to perform the functional rescue therapy on the patient using one or more techniques comprising at least one of: knowledge-based recommendation techniques, machine learning-based recommendation techniques, rule-based expert techniques, network analysis, structural biology integration techniques, gene therapy recommendation techniques, RNA-based therapy generator, drug repurposing module, combination therapy optimizer, personalized dosing calculator, and therapy simulation engine.

[0068] The medical therapy recommendation module **224** is configured to determine the one or more medical therapies for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations. In an embodiment of the present disclosure, the one or more medical therapies correspond to a new drug discovery for one or more potential targets corresponding to the one or more coding variants of the patient. In an exemplary embodiment of the present disclosure, the functional rescue therapy-based AI model is a communication dynamics-based AI model. For example, the one or more medical therapies include an activator therapy, a DNA therapy, an RNA therapy, a protein therapy, a cell therapy, or a combination thereof.

[0069] In an embodiment, for determining the one or more medical therapies for the patient based on the result of the functional rescue therapy, the medical therapy recommendation module **224** with the functional rescue therapy-based AI model is initially configured to obtain at least one of: the result of the functional rescue therapy from the data generation module, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy. The medical therapy recommendation module **224** with the functional rescue therapy-based AI model is further configured to train the functional rescue therapy-based AI model on one or more training datasets associated with one or more known results associated with the functional rescue therapy.

[0070] The medical therapy recommendation module **224** with the functional rescue therapy-based AI model is further configured to extract one or more therapy based features comprising at least one of: molecular signatures of the one or more diseases and therapy response, pathway activation scores, drug-target interaction profiles, and one or more factors corresponding to the patient. The medical therapy recommendation module **224** with the functional rescue therapy-based AI model is further configured to analyze the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy. The medical therapy recommendation module **224** with the functional rescue therapy-based AI model is further configured to rank the one or more medical therapies based on at least one of: predicted efficacy, safety profiles, and the one or more factors corresponding to the patient. The medical therapy recommendation module **224** with the functional rescue therapy-based AI model is further configured to determine the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue therapy-based AI model.

[0071] The data output module **226** is configured to output the generated one or more rescue recommendations and the determined one or more medical therapies on user interface screen of the one or more electronic devices **102** associated with the user.

[0072] In an embodiment of the present disclosure, the drug discovery module **228** generates one or more inhibitor recommendations for the user to perform an inhibitor therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional excess. For generating the one or more inhibitor recommendations, the drug discovery module **228** is configured to: (a) analyze the result of the one or more analysis operations to identify at least one of: specific molecular pathways and proteins exhibiting excess activity; and (b) utilize a knowledge base of known inhibitors and one or more targets of the known inhibitors to generate the one or more inhibitor recommendations.

[0073] Further, the drug discovery module **228** determines if one or more drugs are available for one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations using a target-drug association-based AI model upon generating the one or more inhibitor recommendations. For determining if the one or more drugs are available for one or more drug targets corresponding to the one or more coding variants, the drug discovery module **228** is further configured to: (a) integrate data from at least one of: drug databases, clinical trials, and scientific literature; (b) determine drug-target interactions using one or more machine learning models; and (c) analyze one or more factors comprising at least one of: binding affinity, off-target effects, and drug related properties to determine whether the one or more drugs are available for one or more drug targets.

[0074] Further, the drug discovery module **228** generates one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets. In an embodiment of the present disclosure, the generated one or more drug recommendations are outputted on user interface screen of the one or more electronic devices **102** associated with the user. For generating the one or more drug recommendations for the known drug repositioning corresponding to the determined one or more drugs, the drug discovery module **228** is further configured to: (a) rank the one or more drugs based on the predicted efficacy for one or more drug targets; (b) analyze potential side effects and contraindications; and (c) generate one or more reports indicating rationale for each drug recommendation for the known drug repositioning corresponding to the determined one or more drugs.

[0075] The drug discovery module **228** identifies one or more new drugs for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy upon determining that the one or more drugs are not available for the one or more drug targets. Furthermore, the drug discovery module **228** determines one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy using a structure-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more drug targets. For determining the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets, the drug discovery module **228** with the structure-based drug repurposing engine is configured to: (a) analyze structural similarities between one or more known drug targets and one or more potential drug targets; and (b) determine one or more indications associated with the one or more existing drugs based on the analyzed structural similarities between the one or more known drug targets and the one or more potential drug targets.

[0076] The drug discovery module **228** generates one or more existing drug recommendations for an existing drug repositioning corresponding to the determined one or more existing drugs. In an embodiment of the present disclosure, the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices **102** associated with the user. For generating the one or more existing drug recommendations for the existing drug repositioning, the drug discovery module **228** is further configured to: (a) prioritize the one or more existing drugs based on the predicted efficacy and safety for the one or more indications; (b)

analyze one or more factors comprising at least one of: dosing, formulation, and delivery for use; and (c) generate reports indicating the one or more existing drug recommendations for the existing drug repositioning corresponding to the determined one or more existing drugs based on the analyzed one or more factors.

[0077] Further, the drug repositioning module **230** receives a patient phenotype report of the patient from the one or more electronic devices **102** associated with the user. In an exemplary embodiment of the present disclosure, the patient phenotype report includes height, hair color, DNA sequence of the patient, and the like. The drug repositioning module **230** detects the one or more diseases of the patient based on the patient phenotype report, and International Classification of Diseases (ICD)-9 or ICD-10 using a phenotype-disease association-based AI model. For detecting the one or more diseases of the patient, the drug repositioning module **230** with the phenotype-disease association-based AI model is configured to: (a) obtain the patient phenotype report comprising at least one of: physical characteristics, clinical symptoms and severity, laboratory test results, medical imaging data, DNA sequence from the patient, and the ICD-9 and ICD-10 codes; (b) pre-process the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of numerical data, encoding of categorical variables, tokenization and embedding of text data, image pre-processing and augmentation techniques, and sequence encoding for DNA data; (c) train the phenotype-disease association-based AI model on the one or more training datasets comprising known phenotype-disease associations; (d) extract one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; (e) generate one or more probability scores for the one or more diseases; and (f) detect the one or more diseases for each patient by ranking the one or more diseases based on the generated one or more probability scores.

[0078] Furthermore, the drug repositioning module **230** determines one or more medicines for the detected one or more diseases based on the received patient phenotype report and one or more clinical guidelines using a phenotype-drug association-based AI model upon determining that the detected one or more genome variants are the one or more known variants. For determining the one or more medicines for the detected one or more diseases, the drug repositioning module **230** with the phenotype-drug association-based AI model is configured to: (a) obtain the patient phenotype report comprising at least one of: the clinical symptoms and severity, the laboratory test results, medical history, genetic information, and clinical guidelines relevant to patient condition; (b) pre-process the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of the numerical data, encoding of the categorical variables, and tokenization and embedding of the text data; (c) train the phenotype-drug association-based AI model on the one or more training datasets comprising known phenotype-drug associations; (d) extract the one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; (e) generate the one or more probability scores for the one or more medicines; and (f) detect the one or more medicines for each patient by ranking the one or more medicines based on the generated one or more probability scores.

[0079] The drug repositioning module **230** generates one or more medicine recommendations for a known medicine repositioning corresponding to the determined one or more medicines. For generating the one or more medicine recommendations for the known medicine repositioning corresponding to the determined one or more medicines, the drug repositioning module **230** is configured to: (a) obtain the one or more medicines determined for the detected one or more diseases; (b) analyze the known medicine repositioning by performing one or more repositioning analysis techniques comprising at least one of: target-based screening technique, structure-based analysis, the pathway analysis, side-effect based repositioning technique, and literature mining technique; (c) train the one or more machine learning models on the one or more training datasets comprising one or more successful repositioning cases; (d) predict the known medicine repositioning corresponding to the determined one or more medicines, based on the trained one or

more machine learning modes; and (c) prioritize the one or more medicine recommendations for the known medicine repositioning based on one or more factors comprising at least one of: predicted efficacy, safety profile and potential risks, strength of supporting evidence, and potential market impact and unmet medical needs.

[0080] In an embodiment of the present disclosure, the generated one or more medicine recommendations are outputted on user interface screen of the one or more electronic devices **102** associated with the user.

[0081] In an embodiment of the present disclosure, the data analysis module **232** performs a variant biopathway impact analysis on the extracted patient whole genome report upon determining that the one or more detected more genome variants are the one or more non-coding variants, to assess an impact of the one or more non-coding variants on biological pathways, by: (a) identifying regulatory regions affected by the one or more non-coding variants; (b) predicting one or more changes in transcription factor binding sites; (c) assessing potential impacts on gene expression; (d) mapping affected genes to known biological pathways; and (e) simulating pathway perturbations caused by the one or more non-coding variants. Further, the data analysis module **232** determines if one or more drugs are available for one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a gene-drug association-based AI model upon performing the functional rescue therapy. For determining if the one or more drugs are available for the one or more drug targets corresponding to the one or more non-coding variants, the data analysis module **232** with the gene-drug association-based AI model is configured to: (a) integrate data from one or more drug databases and scientific literature; (b) analyze gene-drug associations using a graph neural network; (c) analyze at least one of: direct and indirect relationship between one or more drug targets; and (d) assess strength and relevance of each potential drug-target association.

[0082] The data analysis module **232** generates one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets. For generating the one or more drug recommendations for the known drug repositioning corresponding to the determined one or more drugs, the data analysis module **232** is configured to: (a) rank drugs based on predicted efficacy for one or more affected pathways; (b) assess potential off-target effects and safety profiles; and (c) generate explanations for each recommendation for the known drug repositioning corresponding to the determined one or more drugs. In an embodiment of the present disclosure, the generated one or more drug recommendations are outputted on user interface screen of the one or more electronic devices **102** associated with the user.

[0083] Furthermore, the data analysis module **232** identifies one or more new drugs for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis upon determining that the one or more drugs are not available for the one or more non-coding variants. For identifying the one or more new drugs for the one or more drug targets corresponding to the one or more non-coding variants, the data analysis module **232** is configured to: (a) perform virtual screening of compound libraries against predicted drug targets; and (b) predict one or more properties and potential efficacy of the one or more new drugs.

[0084] The data analysis module **232** determines one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a pathway-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more non-coding variants. For determining the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants, the data analysis module **232** with the pathway-based drug repurposing engine is configured to: (a) map drug effects onto the biological pathways; (b) identify the one or more existing drugs that compensate for the pathway

perturbations; (c) predict one or more indications based on pathway-level similarities; and (d) analyze drug combinations that synergistically modulate affected pathways.

[0085] The data analysis module **232** further generates one or more existing drug recommendations for a known drug repositioning corresponding to the determined one or more existing drugs. For generating the one or more existing drug recommendations for the known drug repositioning corresponding to the determined one or more existing drugs, the data analysis module **232** is configured to: (a) prioritize the one or more existing drugs based on predicted efficacy and safety; and (b) provide one or more rationales for each recommendation for the known drug repositioning corresponding to the determined one or more existing drugs. In an embodiment of the present disclosure, the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices **102** associated with the user.

[0086] In an embodiment of the present disclosure, the property determination module **234** obtains a molecular structure of one or more biomolecules from the user. In an embodiment of the present disclosure, the molecular structure includes 1-Dimensional (D), 2D, 3D, or a combination thereof. For example, the one or more biomolecules include proteins, and nucleic acids. Further, the property determination module **234** determines one or more properties of each of the one or more biomolecules based on the obtained molecular structure of the one or more biomolecules using a property determination-based AI model. For determining the one or more properties of each of the one or more biomolecules, the property determination module **234** with the property determination-based AI model, is configured to: (a) obtain the molecular structure of the one or more biomolecules from the user; (b) pre-process the molecular structure of the one or more biomolecules to generate pre-processed molecular structure; (c) train the property determination-based AI model on the one or more training datasets comprising one or more known properties; (d) extracting one or more physics-based features comprising at least one of electronic properties and conformational energies, to generate one or more learned embeddings comprising one or more molecular characteristics; (e) processing the molecular structure and sequence data through the trained property determination-based AI model to generate unified representation of the one or more biomolecules; and (f) determining the one or more properties of each of the one or more biomolecules based on the unified representation of the one or more biomolecules. In an embodiment, pre-processing the molecular structure of the one or more biomolecules, comprises: (i) converting the molecular structure into one or more graph representations; (ii) generating molecular fingerprints; (iii) sequence encoding for proteins and nucleic acids; and (iv) normalizing numerical property data. In an exemplary embodiment of the present disclosure, the one or more properties include physical properties, chemical properties, biological properties, or a combination thereof.

[0087] Furthermore, the molecule determination module **236** obtains one or more desired properties of one or more biomolecules from the user. In an exemplary embodiment of the present disclosure, the one or more desired properties include desired physical properties, desired chemical properties, desired biological properties, or a combination thereof. The molecule determination module **236** is configured to determine one or more novel biomolecules based on the obtained one or more desired properties using a molecule determination-based AI model. For determining the one or more novel biomolecules, the molecule determination module **236** with the molecule determination-based AI model, is configured to: (a) obtain the one or more desired properties of the one or more biomolecules from the user; (b) perform a data representation process on the one or more desired properties of the one or more biomolecules; (c) train the molecule determination-based AI model on the one or more training datasets comprising the structure-property relationships using one or more known molecules; (d) generate the one or more biomolecules with the one or more desired properties using the trained molecule determination-based AI model; and (c) process the one or more biomolecules with the one or more desired properties, by at least one of: (i) encoding the one or more desired properties into a latent space representation; (ii) generating

candidate biomolecules based on the one or more encoded properties, using a generative model; (iii) evaluating the candidate molecules using one or more integrated property prediction models; (iv) continuously refining the candidate modules to match the one or more desired properties; and (v) applying one or more chemical validity checks and synthetic accessibility assessments to determine the one or more novel biomolecules.

[0088] In operation, the AI-based computing system **104** creates comprehensive knowledge maps of biological networks relevant to drug-biomarker interactions by integrating machine learning approaches and big data technology. The primary focus of modern biomedical science is to understand molecular level mechanisms and their pathological implications. However, the volume of data produced outstrips current analytical tools. The AI-based computing system **104** constructs comprehensive pathway knowledge maps based on public databases, using the entities (nodes) in databases as keys to perform data mining from research journals and patents. Next, we will develop a fully automated AI system to continually expand and update the maps. Once the map system is generated, probabilistic models can be developed and applied to build predictive models for drug discovery. With this computational resource, it is easier to understand disease models and predict potential drug candidates for potential drug targets. Further, the AI-based computing system **104** develops a cloud-based virtual screening platform for drug candidate screening using conventional computational physics and deep learning models. Machine learning techniques may be used within the framework of computational molecular design to increase accuracy. It may also be used independently as black-box models, trained by experimental data, to increase accuracy and decrease computational cost.

[0089] FIG. **3** is a block diagram depicting an exemplary operation of the AI-based computing system **104**, in accordance with an embodiment of the present disclosure. The drug repositioning module **230** includes a phenotype-disease association engine **302** (i.e., the phenotype-disease association-based AI model), a phenotype-drug association engine **304** (i.e., the phenotype-drug association-based AI model), diagnostics **306**, and a disease-drug association engine **308**. In an embodiment of the present disclosure, the phenotype-disease association engine **302** receives the patient phenotype report **310** and perform the diagnostics **306** to detect the one or more diseases of the patient based on the patient phenotype report, and ICD-9 or ICD-10 using the phenotype-disease association-based AI model. Further, the disease-drug association engine determines the one or more medicines for the detected one or more diseases based on the received patient phenotype report and the one or more clinical guidelines using the phenotype-drug association-based AI model upon determining that the detected one or more genome variants are the one or more known variants. At **312**, the one or more medicine recommendations are generated for the known medicine repositioning corresponding to the determined one or more medicines.

[0090] In an embodiment of the present disclosure, the variant determination module **216** includes a gene/variant-disease association engine **314** configured to determine if the one or more genome variants are the one or more known variants or the one or more unknown variants based on the patient whole genome report **316** and the one or more diseases of the patient using the variant determination-based AI model. At step **318**, it is determined that the one or more genome variants are the one or more known variants. At step **320**, it is determined that the one or more genome variants are the one or more unknown variants. Further, the genome determination module **218** includes a genomic engine **322** configured to determine if the detected one or more genome variants are the one or more coding variants **324** or the one or more non-coding variants **326** based on the patient whole genome report **316** using the coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants. The functional determination module **220** determines if the patient is suffering from the functional loss **328** or the functional excess **330** by performing the one or more analysis operations on the one or more reports upon determining that the detected one or more genome variants are the one or more coding variants **324**.

[0091] In an exemplary embodiment of the present disclosure, the one or more reports are the patient whole genome report **316**, the patient multiomics raw data **332**, the patient RNA report **334**, and the patient exome report **336**. For example, the one or more operations include the expression level check **338** which is performed on the patient RNA report **330**. Further, the one or more operations include the sequence based mutant analysis **340**, the structural based mutant analysis **342**, and the functional impact analysis **344** which are performed on the patient exome report **336**, and the patient multiomics raw data **332**. The data generation module **222** generates the one or more rescue recommendations for the user to perform the functional rescue therapy **346** on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss **328**. Further, the medical therapy recommendation module **224** determines the one or more medical therapies **348** for the patient based on result of the functional rescue therapy **346** using the functional rescue therapy-based AI model upon generating the one or more rescue recommendations.

[0092] The drug discovery module **228** generates the one or more inhibitor recommendations for the user to perform the inhibitor therapy **350** on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional excess **330**. The drug discovery module **228** includes a target association engine **352** determine if the one or more drugs are available for one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations using the target-drug association-based AI model upon generating the one or more inhibitor recommendations. At step **354**, it is determined that the one or more drugs are available and at step **356**, one or more drug recommendations are generated for the known drug repositioning corresponding to the determined one or more drugs. At step **358**, it is determined that the one or more drugs are not available and at step **360**, the one or more new drugs are discovered for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy **350**.

[0093] Further, the drug discovery module **228** includes a structure based drug repositioning engine **362** configured to determine the one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy **350** using the structure-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more drug targets. At step **356**, the one or more existing drug recommendations are generated for an existing drug repositioning corresponding to the determined one or more existing drugs.

[0094] Further, the data analysis module **232** is configured to perform the variant biopathway impact analysis **364** on the extracted patient whole genome report **316** upon determining that the one or more detected genome variants are the one or more non-coding variants **326**. The data analysis module **232** includes a gene/pathway drug association engine **366** configured to determine if one or more drugs are available for one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using the gene-drug association-based AI model upon performing the functional rescue therapy **346**. At step **368**, it is determined that the one or more drugs are available and at step **370**, the one or more drug recommendations are generated for a known drug repositioning corresponding to the determined one or more drugs. At step **372**, it is determined that the one or more drugs are not available and at step **360**, one or more new drugs are identified for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis **364**. The data analysis module **232** includes a pathway-based drug repurposing engine **374** configured to determine one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using the pathway-based drug repurposing engine upon

determining that the one or more drugs are not available for the one or more non-coding variants. At step **370**, one or more existing drug recommendations are generated for a known drug repositioning corresponding to the determined one or more existing drugs.

[0095] FIG. **4** is a block diagram illustrating the AI-based computing system **104** for determining the one or more medical therapies, in accordance with another embodiment of the present disclosure. FIG. **4** depicts that the AI-based computing system **104** includes an application layer **402** that is processed with at least one of: the molecular structure **404** of one or more biomolecules, one or more properties **406** of each of the one or more biomolecules, one or more desired properties **408**, one or more novel biomolecules **410**. The AI-based computing system **104** further includes a computing layer **412** that is configured with at least one of: one or more advanced AI models (GPU/CPU powered AI engine) **414**, a data layer **416**, one or more chemicals **418**, one or more materials **420**, biologics **422**, integrated big data Repo of biomolecules at various scales **424**, and the like, for determining the one or more therapies corresponding to the new drug discovery for the patient.

[0096] FIG. **5** is a process flow diagram illustrating an exemplary AI-based method **500** for determining the one or more medical therapies, in accordance with an embodiment of the present disclosure. At step **502**, the patient multiomics raw data associated with the patient is received from the one or more electronic devices **102** associated with the user or the external database **108**. In an exemplary embodiment of the present disclosure, the user may include the medical professional performing the medical therapy determining process correspond to the drug discovery. In an embodiment of the present disclosure, the drug discovery is the process which aims at identifying the compound therapeutically useful in curing and treating disease. In an exemplary embodiment of the present disclosure, the one or more electronic devices **102** may include a laptop computer, desktop computer, tablet computer, smartphone, wearable device, smart watch, and the like.

[0097] At step **504**, one or more reports are extracted from the received patient multiomics raw data using one or more data extraction techniques. In an exemplary embodiment of the present disclosure, the extracted one or more reports include a patient whole genome report, a patient exome report, and a patient RNA report.

[0098] At step **506**, one or more genome variants are detected in the patient by analyzing the extracted patient whole genome report. In an embodiment of the present disclosure, the one or more genome variants are permanent changes in a DNA sequence that makes up a gene.

[0099] At step **508**, it is determined if the detected one or more genome variants are one or more known variants, or one or more unknown variants based on the extracted patient whole genome report and the one or more diseases of the patient using the variant determination-based AI model.

[0100] At step **510**, it is determined if the detected one or more genome variants are one or more coding variants, or one or more non-coding variants based on the extracted patient whole genome report using a coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants.

[0101] At step **512**, it is determined if the patient is suffering from a functional loss or a functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants. In an embodiment of the present disclosure, the one or more analysis operations include an expression level check, a sequence based mutant analysis, a structural based mutant analysis, a functional impact analysis, or a combination thereof. In an embodiment of the present disclosure, the sequence based mutant analysis, the structure based mutant analysis, and functional impact analysis are performed on the patient exome report, and the expression level check and the functional impact analysis is performed on the patient RNA report to determine if the patient is suffering from the functional loss or the functional excess.

[0102] At step **514**, one or more rescue recommendations are generated for the user to perform the functional rescue therapy on the patient based on result of the one or more analysis operations upon

determining that the patient is suffering from the functional loss.

[0103] At step **516**, the one or more medical therapies are determined for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations. In an embodiment of the present disclosure, the one or more medical therapies correspond to a new drug discovery for one or more potential targets corresponding to the one or more coding variants of the patient. In an embodiment, the one or more medical therapies are determined for the patient using the medical therapy recommendation module **224**, by: (a) obtaining at least one of: the result of the functional rescue therapy from the data generation module **222**, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy; (b) training the functional rescue therapy-based AI model on the one or more training datasets associated with the one or more known results associated with the functional rescue therapy; (c) extracting the one or more therapy based features comprising at least one of: the molecular signatures of the one or more diseases and therapy response, the pathway activation scores, the drug-target interaction profiles, and the one or more factors corresponding to the patient; (d) analyzing the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy; (e) ranking the one or more medical therapies based on at least one of: the predicted efficacy, the safety profiles, and the one or more factors corresponding to the patient; and (f) determining the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue therapy-based AI model. In an exemplary embodiment of the present disclosure, the functional rescue therapy-based AI model is a communication dynamics-based AI model. For example, the one or more medical therapies include an activator therapy, a DNA therapy, an RNA therapy, a protein therapy, a cell therapy, or a combination thereof.

[0104] At step **518**, the generated one or more rescue recommendations and the determined one or more medical therapies are outputted on the user interface screen of the one or more electronic devices **102** associated with the user.

[0105] In an embodiment of the present disclosure, the AI-based method **500** includes generating one or more inhibitor recommendations for the user to perform the inhibitor therapy on the patient based on the result of the one or more analysis operations upon determining that the patient is suffering from the functional excess. Further, the AI-based method **500** includes determining if the one or more drugs are available for the one or more drug targets corresponding to the one or more coding variants based on the result of the one or more analysis operations using the target-drug association-based AI model upon generating the one or more inhibitor recommendations. Further, the AI-based method **500** includes generating the one or more drug recommendations for the known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets. In an embodiment of the present disclosure, the generated one or more drug recommendations are outputted on the user interface screen of the one or more electronic devices **102** associated with the user. The AI-based method **500** includes identifying the one or more new drugs for the one or more drug targets corresponding to the one or more coding variants based on the result of the one or more analysis operations and the inhibitor therapy upon determining that the one or more drugs are not available for the one or more drug targets. Furthermore, the AI-based method **500** includes determining the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets corresponding to the one or more coding variants based on the result of the one or more analysis operations and the inhibitor therapy using the structure-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more drug targets. The AI-based method **500** includes generating the one or more existing drug recommendations for the existing drug repositioning corresponding to the determined one or more existing drugs. In an embodiment of the present disclosure, the generated one or more existing drug recommendations are outputted on the user interface screen of the one or more electronic devices

102 associated with the user.

[0106] Further, the AI-based method **500** includes receiving the patient phenotype report of the patient from the one or more electronic devices **102** associated with the user. In an exemplary embodiment of the present disclosure, the patient phenotype report includes height, hair color, DNA sequence of the patient, and the like. The AI-based method **500** includes detecting the one or more diseases of the patient based on the patient phenotype report, and International Classification of Diseases (ICD)-9 or ICD-10 using the phenotype-disease association-based AI model.

Furthermore, the AI-based method **500** includes determining the one or more medicines for the detected one or more diseases based on the received patient phenotype report and the one or more clinical guidelines using the phenotype-drug association-based AI model upon determining that the detected one or more genome variants are the one or more known variants. The AI-based method **500** includes generating the one or more medicine recommendations for the known medicine repositioning corresponding to the determined one or more medicines. In an embodiment of the present disclosure, the generated one or more medicine recommendations are outputted on the user interface screen of the one or more electronic devices **102** associated with the user.

[0107] In an embodiment of the present disclosure, the AI-based method **500** includes performing the variant biopathway impact analysis on the extracted patient whole genome report upon determining that the one or more detected more genome variants are the one or more non-coding variants. Further, the AI-based method **500** includes determining if one or more drugs are available for one or more drug targets corresponding to the one or more non-coding variants based on the result of the variant biopathway impact analysis using the gene-drug association-based AI model upon performing the functional rescue therapy. The AI-based method **500** includes generating one or more drug recommendations for the known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets. In an embodiment of the present disclosure, the generated one or more drug recommendations are outputted on the user interface screen of the one or more electronic devices **102** associated with the user. Furthermore, the AI-based method **500** includes identifying the one or more new drugs for the one or more drug targets corresponding to the one or more non-coding variants based on the result of the variant biopathway impact analysis upon determining that the one or more drugs are not available for the one or more non-coding variants. The AI-based method **500** includes determining the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using the pathway-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more non-coding variants. The AI-based method **500** includes generating the one or more existing drug recommendations for the known drug repositioning corresponding to the determined one or more existing drugs. In an embodiment of the present disclosure, the generated one or more existing drug recommendations are outputted on the user interface screen of the one or more electronic devices **102** associated with the user.

[0108] In an embodiment of the present disclosure, the AI-based method **500** includes obtaining the molecular structure of the one or more biomolecules from the user. In an embodiment of the present disclosure, the molecular structure includes 1-Dimensional (D), 2D, 3D, or a combination thereof. For example, the one or more biomolecules include proteins, and nucleic acids. Further, the AI-based method **500** includes determining the one or more properties of each of the one or more biomolecules based on the obtained molecular structure of the one or more biomolecules using the property determination-based AI model. In an exemplary embodiment of the present disclosure, the one or more properties include physical properties, chemical properties, biological properties, or a combination thereof.

[0109] Furthermore, the AI-based method **500** includes obtaining the one or more desired properties of one or more biomolecules from the user. In an exemplary embodiment of the present

disclosure, the one or more desired properties include desired physical properties, desired chemical properties, desired biological properties, or a combination thereof. The AI-based method **500** includes determining one or more novel biomolecules based on the obtained one or more desired properties using the molecule determination-based AI model.

[0110] The method **500** may be implemented in any suitable hardware, software, firmware, or combination thereof.

[0111] Thus, various embodiments of the present AI-based computing system **104** provide a solution to facilitate management of drug discovery. In an embodiment of the present disclosure, the AI-based computing system **104** uses communication dynamics powered AI. Thus, the AI-based computing system **104** includes multiple features, such as uniformly applicable, precision adjustable, least action principle for high learning efficiency, first principle, and the like. Further, the AI-based computing system **104** reduces complexity of the biological systems by means of an intelligent engine designed for the drug discovery process. In an embodiment of the present disclosure, a quickened ability to provide new drug treatments to patients would have local and global impact. Furthermore, the AI engine would be generalized internationally. In an embodiment of the present disclosure, the AI-based computing system **104** makes drug development practical by radically shortening development and testing time. The proposed AI-based computing system **104** has the potential to screen **20** billion drug candidates in a few hours.

[0112] The written description describes the subject matter herein to enable any person skilled in the art to make and use the embodiments. The scope of the subject matter embodiments is defined by the claims and may include other modifications that occur to those skilled in the art. Such other modifications are intended to be within the scope of the claims if they have similar elements that do not differ from the literal language of the claims or if they include equivalent elements with insubstantial differences from the literal language of the claims.

[0113] The embodiments herein can comprise hardware and software elements. The embodiments that are implemented in software include but are not limited to, firmware, resident software, microcode, etc. The functions performed by various modules described herein may be implemented in other modules or combinations of other modules. For the purposes of this description, a computer-usable or computer readable medium can be any apparatus that can comprise, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

[0114] The medium can be an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system (or apparatus or device) or a propagation medium. Examples of a computer-readable medium include a semiconductor or solid-state memory, magnetic tape, a removable computer diskette, a random-access memory (RAM), a read-only memory (ROM), a rigid magnetic disk and an optical disk. Current examples of optical disks include compact disk-read only memory (CD-ROM), compact disk-read/write (CD-R/W) and DVD.

[0115] Input/output (I/O) devices (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled to the AI-based computing system **104** either directly or through intervening I/O controllers. Network adapters may also be coupled to the AI-based computing system **104** to enable the data processing system to become coupled to other data processing systems or remote printers or storage devices through intervening private or public networks. Modems, cable modem and Ethernet cards are just a few of the currently available types of network adapters.

[0116] A representative hardware environment for practicing the embodiments may include a hardware configuration of an information handling/AI-based computing system **104** in accordance with the embodiments herein. The AI-based computing system **104** herein comprises at least one processor or central processing unit (CPU). The CPUs are interconnected via system bus **208** to various devices such as a random-access memory (RAM), read-only memory (ROM), and an input/output (I/O) adapter. The I/O adapter can connect to peripheral devices, such as disk units and

tape drives, or other program storage devices that are readable by the AI-based computing system **104**. The AI-based computing system **104** can read the inventive instructions on the program storage devices and follow these instructions to execute the methodology of the embodiments herein.

[0117] The AI-based computing system **104** further includes a user interface adapter that connects a keyboard, mouse, speaker, microphone, and/or other user interface devices such as a touch screen device (not shown) to the bus to gather user input. Additionally, a communication adapter connects the bus to a data processing network, and a display adapter connects the bus to a display device which may be embodied as an output device such as a monitor, printer, or transmitter, for example.

[0118] A description of an embodiment with several components in communication with each other does not imply that all such components are required. On the contrary, a variety of optional components are described to illustrate the wide variety of possible embodiments of the invention.

When a single device or article is described herein, it will be apparent that more than one device/article (whether or not they cooperate) may be used in place of a single device/article. Similarly, where more than one device or article is described herein (whether or not they cooperate), it will be apparent that a single device/article may be used in place of the more than one device or article, or a different number of devices/articles may be used instead of the shown number of devices or programs. The functionality and/or the features of a device may be alternatively embodied by one or more other devices which are not explicitly described as having such functionality/features. Thus, other embodiments of the invention need not include the device itself.

[0119] The illustrated steps are set out to explain the exemplary embodiments shown, and it should be anticipated that ongoing technological development will change the manner in which particular functions are performed. These examples are presented herein for purposes of illustration, and not limitation. Further, the boundaries of the functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternative boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed. Alternatives (including equivalents, extensions, variations, deviations, etc., of those described herein) will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein. Such alternatives fall within the scope and spirit of the disclosed embodiments. Also, the words “comprising,” “having,” “containing,” and “including,” and other similar forms are intended to be equivalent in meaning and be open-ended in that an item or items following any one of these words is not meant to be an exhaustive listing of such item or items or meant to be limited to only the listed item or items. It must also be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0120] Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based here on. Accordingly, the embodiments of the present invention are intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

Claims

1. An Artificial Intelligence (AI)-based computing system for determining medical therapies for a patient, the computing system comprising: one or more hardware processors; and a memory coupled to the one or more hardware processors, wherein the memory comprises a plurality of modules in the form of programmable instructions executable by the one or more hardware processors, and wherein the plurality of modules comprises: a patient data receiver module configured to receive patient multiomics raw data associated with the patient from one of: one or

more electronic devices associated with a user and an external database; a data extracting module configured to extract one or more reports from the received patient multiomics raw data using one or more data extraction techniques, wherein the extracted one or more reports comprise a patient whole genome report, a patient exome report, and a patient Ribonucleic Acid (RNA) report; a variant detection module configured to detect one or more genome variants in the patient by analyzing the extracted patient whole genome report, wherein the one or more genome variants are permanent changes in a Deoxyribonucleic Acid (DNA) sequence that makes up a gene; a variant determination module configured to determine if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants based on the extracted patient whole genome report and one or more diseases of the patient using a variant determination-based Artificial Intelligence (AI) model; a genome determination module configured to determine if the detected one or more genome variants are one of: one or more coding variants and one or more non-coding variants based on the extracted patient whole genome report using a coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants; a functional determination module configured to determine if the patient is suffering from one of: a functional loss and a functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants; a data generation module configured to generate one or more rescue recommendations for the user to perform a functional rescue therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss; a medical therapy recommendation module configured to determine one or more medical therapies for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations, wherein the one or more medical therapies correspond to a new drug discovery for one or more potential targets corresponding to the one or more coding variants of the patient, and wherein in determining the one or more medical therapies for the patient based on the result of the functional rescue therapy, the medical therapy recommendation module with the functional rescue therapy-based AI model, is configured to: obtain at least one of: the result of the functional rescue therapy from the data generation module, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy; train the functional rescue therapy-based AI model on one or more training datasets associated with one or more known results associated with the functional rescue therapy; extract one or more therapy based features comprising at least one of: molecular signatures of the one or more diseases and therapy response, pathway activation scores, drug-target interaction profiles, and one or more factors corresponding to the patient; analyze the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy; rank the one or more medical therapies based on at least one of: predicted efficacy, safety profiles, and the one or more factors corresponding to the patient; and determine the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue therapy-based AI model; and a data output module configured to output the generated one or more rescue recommendations and the determined one or more medical therapies on user interface screen of the one or more electronic devices associated with the user.

2. The AI-based computing system of claim 1, wherein in determining if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants, the variant determination module with the variant determination-based AI model is configured to: obtain information associated with at least one of: the patient whole genome report and the one or more diseases of the patient; extract one or more genome based features from the patient whole genome report, wherein the one or more genome based features comprise at least one of: one or more locations of the one or more genome variants, reference and alternate alleles, and types of the one or more genome variants; train the variant determination-based AI model on the one or more

training datasets comprising the one or more known variants associated with the one or more diseases; compare the one or more known variants against one or more known variant databases to analyze genomic context; correlate the analyzed genomic context with information associated with the one or more diseases of the one or more patient; and classify the one or more genome variants as at least one of: the one or more known variants and the one or more unknown variants, based on the correlation of the analyzed genomic context with the information associated with the one or more diseases of the one or more patient.

3. The AI-based computing system of claim 1, wherein the one or more analysis operations comprises at least one of: expression level check configured to: analyze the patient RNA report to quantify one or more expression levels of genes affected by the one or more coding variants; and compare the one or more expression levels of genes to reference ranges for normal function, wherein the functional loss is determined when the one or more expression levels of genes are below to a pre-determined threshold value, and wherein the functional excess is determined when the one or more expression levels of genes exceed the pre-determined threshold value; sequence-based mutant analysis configured to: determine the patient exome report to analyze specific sequence changes in coding regions; predict an impact of at least one of: amino acid substitutions, insertions, and deletions, on protein function; and utilize computational tools to assess whether mutations are caused to at least one of: disrupt protein activity and optimize the protein activity; structural-based mutant analysis configured to: utilize protein structure prediction tools to model one or more effects of the one or more coding variants on 3D protein structure; analyze an impact of structural changes to at least one of: binding sites, catalytic domains, and overall protein stability; and determine whether structural alterations are caused to at least one of: increase and decrease protein function; functional impact analysis configured to: integrate data from at least one of: the patient RNA report and the exome report, to predict overall functional consequences; analyze one or more factors comprising at least one of: protein domain affected, evolutionary conservation of a mutated region, and known functional effects of similar mutations; and assess whether the variant is predicted to result in at least one of: loss-of-function and gain-of-function effect on the protein; pathway analysis configured to: determine whether the affected proteins interact with biological pathways; determine whether the variant disrupt and over-activate potential cellular processes; and analyze compensatory mechanisms that at least one of: mitigate and exacerbate the functional changes; machine learning-based prediction configured to: utilize one or more trained machine learning models to predict functional outcomes based on features extracted from at least one of: genomic and transcriptomic data; and integrate multiple data types to provide a holistic assessment of functional impact; literature-based analysis configured to: automatically search and analyze scientific literature for reported functional effects of similar variants; and integrate expert knowledge into functional assessment; experimental data integration configured to: integrate data from at least one of: functional assays and animal models studying similar variants; and refine predictions of at least one of: the functional loss and the functional excess using the data.

4. The AI-based computing system of claim 1, wherein the one or more rescue recommendations are generated for the user to perform the functional rescue therapy on the patient using one or more techniques comprising at least one of: knowledge-based recommendation techniques, machine learning-based recommendation techniques, rule-based expert techniques, network analysis, structural biology integration techniques, gene therapy recommendation techniques, RNA-based therapy generator, drug repurposing module, combination therapy optimizer, personalized dosing calculator, and therapy simulation engine.

5. The AI-based computing system of claim 1, further comprising a drug discovery module configured to: generate one or more inhibitor recommendations for the user to perform an inhibitor therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional excess, wherein in generating the one or more inhibitor recommendations, the drug discovery module is configured to: analyze the result of the one or

more analysis operations to identify at least one of: specific molecular pathways and proteins exhibiting excess activity; and utilize a knowledge base of known inhibitors and one or more targets of the known inhibitors to generate the one or more inhibitor recommendations; determine if one or more drugs are available for one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations using a target-drug association-based AI model upon generating the one or more inhibitor recommendations, wherein in determining if the one or more drugs are available for one or more drug targets corresponding to the one or more coding variants, the drug discovery module is further configured to: integrate data from at least one of: drug databases, clinical trials, and scientific literature; determine drug-target interactions using one or more machine learning models; and analyze one or more factors comprising at least one of: binding affinity, off-target effects, and drug related properties to determine whether the one or more drugs are available for one or more drug targets; generate one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets, wherein the generated one or more drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user, wherein in generating the one or more drug recommendations for the known drug repositioning corresponding to the determined one or more drugs, the drug discovery module is further configured to: rank the one or more drugs based on the predicted efficacy for one or more drug targets; analyze potential side effects and contraindications; and generate one or more reports indicating rationale for each drug recommendation for the known drug repositioning corresponding to the determined one or more drugs; identify one or more new drugs for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy upon determining that the one or more drugs are not available for the one or more drug targets; determine one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy using a structure-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more drug targets, wherein in determining the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets, the drug discovery module with the structure-based drug repurposing engine is configured to: analyze structural similarities between one or more known drug targets and one or more potential drug targets; and determine one or more indications associated with the one or more existing drugs based on the analyzed structural similarities between the one or more known drug targets and the one or more potential drug targets; and generate one or more existing drug recommendations for an existing drug repositioning corresponding to the determined one or more existing drugs, wherein the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user, wherein in generating the one or more existing drug recommendations for the existing drug repositioning, the drug discovery module is further configured to: prioritize the one or more existing drugs based on the predicted efficacy and safety for the one or more indications; analyze one or more factors comprising at least one of: dosing, formulation, and delivery for use; and generate reports indicating the one or more existing drug recommendations for the existing drug repositioning corresponding to the determined one or more existing drugs based on the analyzed one or more factors.

6. The AI-based computing system of claim 1, further comprising a drug repositioning module configured to: receive a patient phenotype report of the patient from the one or more electronic devices associated with the user, wherein the patient phenotype report comprises height, hair color, and DNA sequence of the patient; detect one or more diseases of the patient based on the patient phenotype report and one of: International Classification of Diseases (ICD)-9 and ICD-10 using a phenotype-disease association-based AI model, by: obtaining the patient phenotype report

comprising at least one of: physical characteristics, clinical symptoms and severity, laboratory test results, medical imaging data, DNA sequence from the patient, and the ICD-9 and ICD-10 codes; pre-processing the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of numerical data, encoding of categorical variables, tokenization and embedding of text data, image pre-processing and augmentation techniques, and sequence encoding for DNA data; training the phenotype-disease association-based AI model on the one or more training datasets comprising known phenotype-disease associations; extracting one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; generating one or more probability scores for the one or more diseases; and detecting the one or more diseases for each patient by ranking the one or more diseases based on the generated one or more probability scores; determine one or more medicines for the detected one or more diseases based on the received patient phenotype report and one or more clinical guidelines using a phenotype-drug association-based AI model upon determining that the detected one or more genome variants are the one or more known variants, by: obtaining the patient phenotype report comprising at least one of: the clinical symptoms and severity, the laboratory test results, medical history, genetic information, and clinical guidelines relevant to patient condition; pre-processing the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of the numerical data, encoding of the categorical variables, and tokenization and embedding of the text data; training the phenotype-drug association-based AI model on the one or more training datasets comprising known phenotype-drug associations; extracting the one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; generating the one or more probability scores for the one or more medicines; and detecting the one or more medicines for each patient by ranking the one or more medicines based on the generated one or more probability scores; and generate one or more medicine recommendations for a known medicine repositioning corresponding to the determined one or more medicines, by: obtaining the one or more medicines determined for the detected one or more diseases; analyzing the known medicine repositioning by performing one or more repositioning analysis techniques comprising at least one of: target-based screening technique, structure-based analysis, the pathway analysis, side-effect based repositioning technique, and literature mining technique; training the one or more machine learning models on the one or more training datasets comprising one or more successful repositioning cases; predicting the known medicine repositioning corresponding to the determined one or more medicines, based on the trained one or more machine learning modes; and prioritizing the one or more medicine recommendations for the known medicine repositioning based on one or more factors comprising at least one of: predicted efficacy, safety profile and potential risks, strength of supporting evidence, and potential market impact and unmet medical needs, wherein the generated one or more medicine recommendations are outputted on user interface screen of the one or more electronic devices associated with the user.

7. The AI-based computing system of claim 1, wherein the data analysis module is configured to: perform a variant biopathway impact analysis on the extracted patient whole genome report upon determining that the one or more detected more genome variants are the one or more non-coding variants, to assess an impact of the one or more non-coding variants on biological pathways, by: identifying regulatory regions affected by the one or more non-coding variants; predicting one or more changes in transcription factor binding sites; assessing potential impacts on gene expression; mapping affected genes to known biological pathways; and simulating pathway perturbations caused by the one or more non-coding variants; determine if one or more drugs are available for one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a gene-drug association-based AI model upon performing the functional rescue therapy, by: integrating data from one or more drug databases and scientific literature; analyzing gene-drug associations using a graph neural network; analyzing at

least one of: direct and indirect relationship between one or more drug targets; and assessing strength and relevance of each potential drug-target association; generate one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets, by: ranking drugs based on predicted efficacy for one or more affected pathways; assessing potential off-target effects and safety profiles; and generating explanations for each recommendation for the known drug repositioning corresponding to the determined one or more drugs, wherein the generated one or more drug recommendations are outputted in one or more formats on user interface screen of the one or more electronic devices associated with the user; identify one or more new drugs for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis upon determining that the one or more drugs are not available for the one or more non-coding variants, by: performing virtual screening of compound libraries against predicted drug targets; and predicting one or more properties and potential efficacy of the one or more new drugs; determine one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a pathway-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more non-coding variants, by: mapping drug effects onto the biological pathways; identifying the one or more existing drugs that compensate for the pathway perturbations; predicting one or more indications based on pathway-level similarities; and analyzing drug combinations that synergistically modulate affected pathways; and generate one or more existing drug recommendations for a known drug repositioning corresponding to the determined one or more existing drugs, by: prioritizing the one or more existing drugs based on predicted efficacy and safety; and providing one or more rationales for each recommendation for the known drug repositioning corresponding to the determined one or more existing drugs, wherein the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user.

8. The AI-based computing system of claim 1, further comprising a property determination module configured to: obtain a molecular structure of one or more biomolecules from the user, wherein the molecular structure comprises at least one of: 1-Dimensional (D), 2D, and 3D, and wherein the one or more biomolecules comprise proteins, and nucleic acids; and determine one or more properties of each of the one or more biomolecules based on the obtained molecular structure of the one or more biomolecules using a property determination-based AI model, by: obtaining the molecular structure of the one or more biomolecules from the user; pre-processing the molecular structure of the one or more biomolecules to generate pre-processed molecular structure, by: converting the molecular structure into one or more graph representations; generating molecular fingerprints; sequence encoding for proteins and nucleic acids; and normalizing numerical property data; training the property determination-based AI model on the one or more training datasets comprising one or more known properties; extracting one or more physics-based features comprising at least one of electronic properties and conformational energies, to generate one or more learned embeddings comprising one or more molecular characteristics; processing the molecular structure and sequence data through the trained property determination-based AI model to generate unified representation of the one or more biomolecules; and determining the one or more properties of each of the one or more biomolecules based on the unified representation of the one or more biomolecules, wherein the one or more properties comprise at least one of: physical properties, chemical properties and biological properties.

9. The AI-based computing system of claim 1, further comprising a molecule determination module configured to: obtain one or more desired properties of one or more biomolecules from the user, wherein the one or more desired properties comprise at least one of: desired physical properties, desired chemical properties and desired biological properties; and determine one or more novel

biomolecules based on the obtained one or more desired properties using a molecule determination-based AI model, by: obtaining the one or more desired properties of the one or more biomolecules from the user; performing a data representation process on the one or more desired properties of the one or more biomolecules; training the molecule determination-based AI model on the one or more training datasets comprising the structure-property relationships using one or more known molecules; generating the one or more biomolecules with the one or more desired properties using the trained molecule determination-based AI model; and processing the one or more biomolecules with the one or more desired properties, by at least one of: encoding the one or more desired properties into a latent space representation; generating candidate biomolecules based on the one or more encoded properties, using a generative model; evaluating the candidate molecules using one or more integrated property prediction models; continuously refining the candidate modules to match the one or more desired properties; and applying one or more chemical validity checks and synthetic accessibility assessments to determine the one or more novel biomolecules.

10. An Artificial Intelligence (AI)-based method for determining medical therapies for a patient, the method comprising: receiving, by one or more hardware processors, patient multiomics raw data associated with the patient from one of: one or more electronic devices associated with a user and an external database; extracting, by the one or more hardware processors, one or more reports from the received patient multiomics raw data using one or more data extraction techniques, wherein the extracted one or more reports comprise a patient whole genome report, a patient exome report, and a patient Ribonucleic Acid (RNA) report; detecting, by the one or more hardware processors, one or more genome variants in the patient by analyzing the extracted patient whole genome report, wherein the one or more genome variants are permanent changes in a Deoxyribonucleic Acid (DNA) sequence that makes up a gene; determining, by the one or more hardware processors, if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants based on the extracted patient whole genome report and one or more diseases of the patient using a variant determination-based Artificial Intelligence (AI) model; determining, by the one or more hardware processors, if the detected one or more genome variants are one of: one or more coding variants and one or more non-coding variants based on the extracted patient whole genome report using a coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants; determining, by one or more hardware processors, if the patient is suffering from one of: a functional loss and a functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants; generating, by one or more hardware processors, one or more rescue recommendations for the user to perform a functional rescue therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss; determining, by the one or more hardware processors, one or more medical therapies for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations, wherein the one or more medical therapies correspond to a new drug discovery for one or more potential targets corresponding to the one or more coding variants of the patient, and wherein determining the one or more medical therapies for the patient based on the result of the functional rescue therapy, comprises: obtaining, by the one or more hardware processors, at least one of: the result of the functional rescue therapy from the data generation module, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy; training, by the one or more hardware processors, the functional rescue therapy-based AI model on one or more training datasets associated with one or more known results associated with the functional rescue therapy; extracting, by the one or more hardware processors, one or more therapy based features comprising at least one of: molecular signatures of the one or more diseases and therapy response, pathway activation scores, drug-target interaction profiles, and one or more factors corresponding to the patient; analyzing, by the one or more

hardware processors, the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy; ranking, by the one or more hardware processors, the one or more medical therapies based on at least one of: predicted efficacy, safety profiles, and the one or more factors corresponding to the patient; and determining, by the one or more hardware processors, the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue therapy-based AI model; and outputting, by the one or more hardware processors, the generated one or more rescue recommendations and the determined one or more medical therapies on user interface screen of the one or more electronic devices associated with the user.

11. The AI-based computing method of claim 10, wherein determining if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants, comprises: obtaining, by the one or more hardware processors, information associated with at least one of: the patient whole genome report and the one or more diseases of the patient; extracting, by the one or more hardware processors, one or more genome based features from the patient whole genome report, wherein the one or more genome based features comprise at least one of: one or more locations of the one or more genome variants, reference and alternate alleles, and types of the one or more genome variants; training, by the one or more hardware processors, the variant determination-based AI model on the one or more training datasets comprising the one or more known variants associated with the one or more diseases; comparing, by the one or more hardware processors, the one or more known variants against one or more known variant databases to analyze genomic context; correlating, by the one or more hardware processors, the analyzed genomic context with information associated with the one or more diseases of the one or more patient; and classifying, by the one or more hardware processors, the one or more genome variants as at least one of: the one or more known variants and the one or more unknown variants, based on the correlation of the analyzed genomic context with the information associated with the one or more diseases of the one or more patient.

12. The AI-based computing method of claim 10, wherein the one or more analysis operations comprises at least one of: expression level check that performs: analyzing, by the one or more hardware processors, the patient RNA report to quantify one or more expression levels of genes affected by the one or more coding variants; and comparing, by the one or more hardware processors, the one or more expression levels of genes to reference ranges for normal function, wherein the functional loss is determined when the one or more expression levels of genes are below to a pre-determined threshold value, and wherein the functional excess is determined when the one or more expression levels of genes exceed the pre-determined threshold value; sequence-based mutant analysis that performs: determining, by the one or more hardware processors, the patient exome report to analyze specific sequence changes in coding regions; predicting, by the one or more hardware processors, an impact of at least one of: amino acid substitutions, insertions, and deletions, on protein function; and utilizing, by the one or more hardware processors, computational tools to assess whether mutations are caused to at least one of: disrupt protein activity and optimize the protein activity; structural-based mutant analysis that performs: utilizing, by the one or more hardware processors, protein structure prediction tools to model one or more effects of the one or more coding variants on 3D protein structure; analyzing, by the one or more hardware processors, an impact of structural changes to at least one of: binding sites, catalytic domains, and overall protein stability; and determining, by the one or more hardware processors, whether structural alterations are caused to at least one of: increase and decrease protein function; functional impact analysis that performs: integrating, by the one or more hardware processors, data from at least one of: the patient RNA report and the exome report, to predict overall functional consequences; analyzing, by the one or more hardware processors, one or more factors comprising at least one of: protein domain affected, evolutionary conservation of a mutated region, and known functional effects of similar mutations; and assessing, by the one or more hardware processors,

whether the variant is predicted to result in at least one of: loss-of-function and gain-of-function effect on the protein; pathway analysis that performs determining, by the one or more hardware processors, whether the affected proteins interact with biological pathways; determining, by the one or more hardware processors, whether the variant disrupt and over-activate potential cellular processes; and analyzing, by the one or more hardware processors, compensatory mechanisms that at least one of: mitigate and exacerbate the functional changes; machine learning-based prediction configured to: utilizing, by the one or more hardware processors, one or more trained machine learning models to predict functional outcomes based on features extracted from at least one of: genomic and transcriptomic data; and integrating, by the one or more hardware processors, multiple data types to provide a holistic assessment of functional impact; literature-based analysis that performs: automatically searching and analyzing, by the one or more hardware processors, scientific literature for reported functional effects of similar variants; and integrating, by the one or more hardware processors, expert knowledge into functional assessment; experimental data integration that performs: integrating, by the one or more hardware processors, data from at least one of: functional assays and animal models studying similar variants; and refining, by the one or more hardware processors, predictions of at least one of: the functional loss and the functional excess using the data.

13. The AI-based computing method of claim 10, wherein the one or more rescue recommendations are generated for the user to perform the functional rescue therapy on the patient using one or more techniques comprising at least one of: knowledge-based recommendation techniques, machine learning-based recommendation techniques, rule-based expert techniques, network analysis, structural biology integration techniques, gene therapy recommendation techniques, RNA-based therapy generator, drug repurposing module, combination therapy optimizer, personalized dosing calculator, and therapy simulation engine.

14. The AI-based method of claim 10, further comprising: generating, by the one or more hardware processors, one or more inhibitor recommendations for the user to perform an inhibitor therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional excess, wherein generating the one or more inhibitor recommendations, comprises: analyzing, by the one or more hardware processors, the result of the one or more analysis operations to identify at least one of: specific molecular pathways and proteins exhibiting excess activity; utilizing, by the one or more hardware processors, a knowledge base of known inhibitors and one or more targets of the known inhibitors to generate the one or more inhibitor recommendations; determining if one or more drugs are available for one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations using a target-drug association-based AI model upon generating the one or more inhibitor recommendations, wherein determining if the one or more drugs are available for one or more drug targets corresponding to the one or more coding variants, comprises: integrating, by the one or more hardware processors, data from at least one of: drug databases, clinical trials, and scientific literature; determining, by the one or more hardware processors, drug-target interactions using one or more machine learning models; and analyzing, by the one or more hardware processors, one or more factors comprising at least one of: binding affinity, off-target effects, and drug related properties to determine whether the one or more drugs are available for one or more drug targets; generating one or more drug recommendations for a known drug repositioning corresponding to the determined one or more drugs upon determining that the one or more drugs are available for the one or more drug targets, wherein the generated one or more drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user, wherein in generating the one or more drug recommendations for the known drug repositioning corresponding to the determined one or more drugs, comprises: ranking, by the one or more hardware processors, the one or more drugs based on the predicted efficacy for one or more drug targets; analyzing, by the one or more hardware processors, potential side effects

and contraindications; and generating, by the one or more hardware processors, one or more reports indicating rationale for each drug recommendation for the known drug repositioning corresponding to the determined one or more drugs; identifying one or more new drugs for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy upon determining that the one or more drugs are not available for the one or more drug targets; determining one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more coding variants based on result of the one or more analysis operations and the inhibitor therapy using a structure-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more drug targets, wherein in determining the one or more existing drugs having the ability to provide the one or more therapeutic uses for the one or more drug targets, comprises: analyzing, by the one or more hardware processors, structural similarities between one or more known drug targets and one or more potential drug targets; and determining, by the one or more hardware processors, one or more indications associated with the one or more existing drugs based on the analyzed structural similarities between the one or more known drug targets and the one or more potential drug targets; and generating one or more existing drug recommendations for an existing drug repositioning corresponding to the determined one or more existing drugs, wherein the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user, wherein generating the one or more existing drug recommendations for the existing drug repositioning, the drug discovery module is further configured to: prioritizing, by the one or more hardware processors, the one or more existing drugs based on the predicted efficacy and safety for the one or more indications; analyzing, by the one or more hardware processors, one or more factors comprising at least one of: dosing, formulation, and delivery for use; and generating, by the one or more hardware processors, reports indicating the one or more existing drug recommendations for the existing drug repositioning corresponding to the determined one or more existing drugs based on the analyzed one or more factors.

15. The AI-based method of claim 10, further comprising: receiving, by the one or more hardware processors, a patient phenotype report of the patient from the one or more electronic devices associated with the user, wherein the patient phenotype report comprises height, hair color, and DNA sequence of the patient; detecting, by the one or more hardware processors, one or more diseases of the patient based on the patient phenotype report and one of: International Classification of Diseases (ICD)-9 and ICD-10 using a phenotype-disease association-based AI model, by: obtaining, by the one or more hardware processors, the patient phenotype report comprising at least one of: physical characteristics, clinical symptoms and severity, laboratory test results, medical imaging data, DNA sequence from the patient, and the ICD-9 and ICD-10 codes; pre-processing, by the one or more hardware processors, the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of numerical data, encoding of categorical variables, tokenization and embedding of text data, image pre-processing and augmentation techniques, and sequence encoding for DNA data; training, by the one or more hardware processors, the phenotype-disease association-based AI model on the one or more training datasets comprising known phenotype-disease associations; extracting, by the one or more hardware processors, one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; generating, by the one or more hardware processors, one or more probability scores for the one or more diseases; and detecting, by the one or more hardware processors, the one or more diseases for each patient by ranking the one or more diseases based on the generated one or more probability scores; determining, by the one or more hardware processors, one or more medicines for the detected one or more diseases based on the received patient phenotype report and one or more clinical guidelines using a phenotype-drug association-based AI model upon determining that the detected one or more genome variants are the one or

more known variants, by: obtaining, by the one or more hardware processors, the patient phenotype report comprising at least one of: the clinical symptoms and severity, the laboratory test results, medical history, genetic information, and clinical guidelines relevant to patient condition; pre-processing, by the one or more hardware processors, the patient phenotype report to generate pre-processed data, wherein pre-processing comprises at least one of: normalization and standardization of the numerical data, encoding of the categorical variables, and tokenization and embedding of the text data; training, by the one or more hardware processors, the phenotype-drug association-based AI model on the one or more training datasets comprising known phenotype-drug associations; extracting, by the one or more hardware processors, the one or more phenotypic based features to identify one or more phenotypic traits for each potential disease; generating, by the one or more hardware processors, the one or more probability scores for the one or more medicines; and detecting, by the one or more hardware processors, the one or more medicines for each patient by ranking the one or more medicines based on the generated one or more probability scores; and generating, by the one or more hardware processors, one or more medicine recommendations for a known medicine repositioning corresponding to the determined one or more medicines, by: obtaining, by the one or more hardware processors, the one or more medicines determined for the detected one or more diseases; analyzing, by the one or more hardware processors, the known medicine repositioning by performing one or more repositioning analysis techniques comprising at least one of: target-based screening technique, structure-based analysis, the pathway analysis, side-effect based repositioning technique, and literature mining technique; training, by the one or more hardware processors, the one or more machine learning models on the one or more training datasets comprising one or more successful repositioning cases; predicting, by the one or more hardware processors, the known medicine repositioning corresponding to the determined one or more medicines, based on the trained one or more machine learning modes; and prioritizing, by the one or more hardware processors, the one or more medicine recommendations for the known medicine repositioning based on one or more factors comprising at least one of: predicted efficacy, safety profile and potential risks, strength of supporting evidence, and potential market impact and unmet medical needs, wherein the generated one or more medicine recommendations are outputted on user interface screen of the one or more electronic devices associated with the user.

16. The AI-based method of claim 10, further comprising: performing, by the one or more hardware processors, a variant biopathway impact analysis on the extracted patient whole genome report upon determining that the one or more detected more genome variants are the one or more non-coding variants, to assess an impact of the one or more non-coding variants on biological pathways, by: identifying, by the one or more hardware processors, regulatory regions affected by the one or more non-coding variants; predicting, by the one or more hardware processors, one or more changes in transcription factor binding sites; assessing, by the one or more hardware processors, potential impacts on gene expression; mapping, by the one or more hardware processors, affected genes to known biological pathways; and simulating, by the one or more hardware processors, pathway perturbations caused by the one or more non-coding variants; determining, by the one or more hardware processors, if one or more drugs are available for one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a gene-drug association-based AI model upon performing the functional rescue therapy, by: integrating, by the one or more hardware processors, data from one or more drug databases and scientific literature; analyzing, by the one or more hardware processors, gene-drug associations using a graph neural network; analyzing, by the one or more hardware processors, at least one of: direct and indirect relationship between one or more drug targets; and assessing, by the one or more hardware processors, strength and relevance of each potential drug-target association; generating, by the one or more hardware processors, one or more drug recommendations for a known drug repositioning corresponding to the determined one or

more drugs upon determining that the one or more drugs are available for the one or more drug targets, by: ranking, by the one or more hardware processors, drugs based on predicted efficacy for one or more affected pathways; assessing, by the one or more hardware processors, potential off-target effects and safety profiles; and generating, by the one or more hardware processors, explanations for each recommendation for the known drug repositioning corresponding to the determined one or more drugs, wherein the generated one or more drug recommendations are outputted in one or more formats on user interface screen of the one or more electronic devices associated with the user; identifying, by the one or more hardware processors, one or more new drugs for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis upon determining that the one or more drugs are not available for the one or more non-coding variants, by: performing, by the one or more hardware processors, virtual screening of compound libraries against predicted drug targets; and predicting, by the one or more hardware processors, one or more properties and potential efficacy of the one or more new drugs; determining, by the one or more hardware processors, one or more existing drugs having an ability to provide one or more therapeutic uses for the one or more drug targets corresponding to the one or more non-coding variants based on result of the variant biopathway impact analysis using a pathway-based drug repurposing engine upon determining that the one or more drugs are not available for the one or more non-coding variants, by: mapping, by the one or more hardware processors, drug effects onto the biological pathways; identifying, by the one or more hardware processors, the one or more existing drugs that compensate for the pathway perturbations; predicting, by the one or more hardware processors, one or more indications based on pathway-level similarities; and analyzing, by the one or more hardware processors, drug combinations that synergistically modulate affected pathways; and generating, by the one or more hardware processors, one or more existing drug recommendations for a known drug repositioning corresponding to the determined one or more existing drugs, by: prioritizing, by the one or more hardware processors, the one or more existing drugs based on predicted efficacy and safety; and providing, by the one or more hardware processors, one or more rationales for each recommendation for the known drug repositioning corresponding to the determined one or more existing drugs, wherein the generated one or more existing drug recommendations are outputted on user interface screen of the one or more electronic devices associated with the user.

17. The AI-based method of claim 10, further comprising: obtaining, by the one or more hardware processors, a molecular structure of one or more biomolecules from the user, wherein the molecular structure comprises at least one of: 1-Dimensional (D), 2D, and 3D, and wherein the one or more biomolecules comprise proteins, and nucleic acids; and determining, by the one or more hardware processors, one or more properties of each of the one or more biomolecules based on the obtained molecular structure of the one or more biomolecules using a property determination-based AI model, by: obtaining, by the one or more hardware processors, the molecular structure of the one or more biomolecules from the user; pre-processing, by the one or more hardware processors, the molecular structure of the one or more biomolecules to generate pre-processed molecular structure, by: converting, by the one or more hardware processors, the molecular structure into one or more graph representations; generating, by the one or more hardware processors, molecular fingerprints; sequence, by the one or more hardware processors, encoding for proteins and nucleic acids; and normalizing, by the one or more hardware processors, numerical property data; training, by the one or more hardware processors, the property determination-based AI model on the one or more training datasets comprising one or more known properties; extracting, by the one or more hardware processors, one or more physics-based features comprising at least one of electronic properties and conformational energies, to generate one or more learned embeddings comprising one or more molecular characteristics; processing, by the one or more hardware processors, the molecular structure and sequence data through the trained property determination-based AI model to generate unified representation of the one or more biomolecules; and determining, by the one or

more hardware processors, the one or more properties of each of the one or more biomolecules based on the unified representation of the one or more biomolecules, wherein the one or more properties comprise at least one of: physical properties, chemical properties and biological properties.

18. The AI-based method of claim 10, further comprising: obtaining, by the one or more hardware processors, one or more desired properties of one or more biomolecules from the user, wherein the one or more desired properties comprise at least one of: desired physical properties, desired chemical properties and desired biological properties; and determining, by the one or more hardware processors, one or more novel biomolecules based on the obtained one or more desired properties using a molecule determination-based AI model, by: obtaining, by the one or more hardware processors, the one or more desired properties of the one or more biomolecules from the user; performing, by the one or more hardware processors, a data representation process on the one or more desired properties of the one or more biomolecules; training, by the one or more hardware processors, the molecule determination-based AI model on the one or more training datasets comprising the structure-property relationships using one or more known molecules; generating, by the one or more hardware processors, the one or more biomolecules with the one or more desired properties using the trained molecule determination-based AI model; and processing, by the one or more hardware processors, the one or more biomolecules with the one or more desired properties, by at least one of: encoding, by the one or more hardware processors, the one or more desired properties into a latent space representation; generating, by the one or more hardware processors, candidate biomolecules based on the one or more encoded properties, using a generative model; evaluating, by the one or more hardware processors, the candidate molecules using one or more integrated property prediction models; continuously refining, by the one or more hardware processors, the candidate modules to match the one or more desired properties; and applying, by the one or more hardware processors, one or more chemical validity checks and synthetic accessibility assessments to determine the one or more novel biomolecules.

19. A non-transitory computer-readable storage medium having instructions stored therein that, when executed by a hardware processor, cause the processor to perform method steps comprising: receiving patient multiomics raw data associated with a patient from one of: one or more electronic devices associated with a user and an external database; extracting one or more reports from the received patient multiomics raw data using one or more data extraction techniques, wherein the extracted one or more reports comprise a patient whole genome report, a patient exome report, and a patient Ribonucleic Acid (RNA) report; detecting one or more genome variants in the patient by analyzing the extracted patient whole genome report, wherein the one or more genome variants are permanent changes in a Deoxyribonucleic Acid (DNA) sequence that makes up a gene; determining if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants based on the extracted patient whole genome report and one or more diseases of the patient using a variant determination-based Artificial Intelligence (AI) model; determining if the detected one or more genome variants are one of: one or more coding variants and one or more non-coding variants based on the extracted patient whole genome report using a coding determination-based AI model upon determining that the detected one or more genome variants are the one or more unknown variants; determining if the patient is suffering from one of: a functional loss and a functional excess by performing one or more analysis operations on the extracted one or more reports upon determining that the detected one or more genome variants are the one or more coding variants; generating one or more rescue recommendations for the user to perform a functional rescue therapy on the patient based on result of the one or more analysis operations upon determining that the patient is suffering from the functional loss; determining one or more medical therapies for the patient based on result of the functional rescue therapy using a functional rescue therapy-based AI model upon generating the one or more rescue recommendations, wherein the one or more medical therapies correspond to a new drug discovery

for one or more potential targets corresponding to the one or more coding variants of the patient, and wherein determining the one or more medical therapies for the patient based on the result of the functional rescue therapy, comprises: obtaining at least one of: the result of the functional rescue therapy from the data generation module, genomic and clinical data of the patient, and historical data associated with the functional rescue therapy; training the functional rescue therapy-based AI model on one or more training datasets associated with one or more known results associated with the functional rescue therapy; extracting one or more therapy based features comprising at least one of: molecular signatures of the one or more diseases and therapy response, pathway activation scores, drug-target interaction profiles, and one or more factors corresponding to the patient; analyzing the result of the functional rescue therapy to identify at least one of: functional deficits and unexpected effects of the functional rescue therapy; ranking the one or more medical therapies based on at least one of: predicted efficacy, safety profiles, and the one or more factors corresponding to the patient; and determining the one or more medical therapies for the patient based on the ranking of the one or more medical therapies using the functional rescue therapy-based AI model; and outputting the generated one or more rescue recommendations and the determined one or more medical therapies on user interface screen of the one or more electronic devices associated with the user.

20. The non-transitory computer-readable storage medium of claim 19, wherein determining if the detected one or more genome variants are one of: one or more known variants and one or more unknown variants, comprises: obtaining information associated with at least one of: the patient whole genome report and the one or more diseases of the patient; extracting one or more genome based features from the patient whole genome report, wherein the one or more genome based features comprise at least one of: one or more locations of the one or more genome variants, reference and alternate alleles, and types of the one or more genome variants; training the variant determination-based AI model on the one or more training datasets comprising the one or more known variants associated with the one or more diseases; comparing the one or more known variants against one or more known variant databases to analyze genomic context; correlating the analyzed genomic context with information associated with the one or more diseases of the one or more patient; and classifying the one or more genome variants as at least one of: the one or more known variants and the one or more unknown variants, based on the correlation of the analyzed genomic context with the information associated with the one or more diseases of the one or more patient.
